ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

Master of Computer Applications(M.C.A) Programme

(Choice Based Credit System)

ACADEMIC RULES

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. *Duration of Programme*

Postgraduate programme: Master of Computer Applications

Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

3. *Eligibility & Mode of admissions*

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. *Programme structure and Credits*

A student admitted to a program should study the course and earn credits specified in the course structure.

5. *Attendance*

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6 *Course Evaluation*

6.1 The performance of every student in each course will be evaluated as follows:

6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 ***University Examination***

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3 Performance at Internal & University Examination will be done on the relative grading system.

7 Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

SGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses in the semester

CGPA = $\Sigma C_i G_i / \Sigma C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

8. Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in course structure.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥7.5 & ≤10
First class:	CGPA ≥ 6.0 & <7.5
Second Class:	CGPA ≥ 5.0 & <6.0
Pass Class:	CGPA <5.0

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates

proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision,Mission,PEOs,POs and PSOs

Vision:

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will

PEO1: Be able to understand and analyse the requirements of complex real world problems and solve them by designing optimised computer applications.

PEO2: Be able to apply skills in order to excel at professional careers.

PEO3: Be able adopt at rapidly changing technological landscape in domains of their preference.

PEO4: Be able to act ethically and professionally while maintaining high moral values.

PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to:

PO1. Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO2. Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO3. Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies
PO4. Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.
PO5. Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions
PO6. Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO7. **Life-long Learning**: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO8. Project Management and Finance: Ability to understand, management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO9. Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.

PO10. **Societal & Environmental Concern:** Ability to recognize economical, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11. Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO12. Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the programme, the student should be able to

PSO1: Analyse, design and develop effective software applications for solving contemporary challenges.

PSO2: Harness the necessary skills with spirit of research and entrepreneurship.

Total Credits of Programme(4 Semesters) **108**

TEACHING SCHEME
FOR

MCA PROGRAMME

EFFECTIVE FROM
ACADEMIC YEAR 2022-23

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA859-862	Elective-I	4	-	4	4	10	20	70	-	-	-	100
CA863	Enterprise Computing using Java EE	4	3	7	7	10	20	70	15	15	70	200
CA864	Programming with .NET Architecture	3	3	6	6	10	20	70	15	15	70	200
CA865	NoSQL Database	4	3	7	6	10	20	70	15	15	70	200
HS141.02 C	Foreign Languages	-	2	2	2	-		-	30		70	100
HS105.02 C	Academic Speaking and Presentation Skills					-						
		15	11	26	26	400			400			800

Elective-I	
Course Code	Course Title
CA859	Cloud Computing
CA860	Cryptography & Network Security
CA861	Block Chain Technology
CA862	Quantum Computing

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA866-869	Elective-II	3	3	6	6	10	20	70	15	15	70	200
CA870	Software Quality Assurance	3	3	6	6	10	20	70	15	15	70	200
CA871	Advanced Mobile Programming	3	3	6	6	10	20	70	15	15	70	200
CA872	Software Engineering with Agile and DevOps	4	-	4	4	10	20	70	-	-	-	100
HS106.02 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-III **	-	2	2	2	-		-	30		70	100
		13	13	26	26	400			500			900

Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Internet and Web Designing and Mobile Application Development** course for others.

Elective-II	
Course Code	Course Title
CA866	Game Development using Unity
CA867	Python Web framework
CA868	Framework and Applications
CA869	HTTP Web Service for Enterprise Application

University Elective-I

No	Course Code	Course Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	MB651	Software based Statistical Analysis	Management
5	NR755	First Aid & Life Support	Nursing
6	OC733.01	Introduction to Polymer Science	Applied Science
7	MA771.01	Reliability and Risk Analysis	Mathematics
8	ME781.01	Occupational Health &Safety	Engineering
9	MA772.01	Design of Experiments	Mathematics
10	RD701.01	Introduction to Analytical Techniques	Applied Science

11	RD702.01	Introduction to Nanoscience And Technology	Applied Science
12	PH891	Community Pharmacy Ownership	Pharmacy
13	PH892	Intellectual Property Rights	Pharmacy
14	PSE55	Astrophysics ,Space and Cosmos	Applied Science

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA934-937	Elective-III	3	3	6	6	10	20	70	15	15	70	200
CA938	Advanced Web Designing	3	3	6	6	10	20	70	15	15	70	200
CA939	Mini Project	-	12	12	12	-	-	-	100		300	400
CA940	Green Computing		2	2	02	-	-	-	30		70	100
		06	20	26	26	200			700			900

Elective-III	
Course Code	Course Title
CA934	Digital Image Processing
CA935	Object Oriented Programming for Smart Contracts
CA936	Data Analytics
CA937	Internet of Things

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester-IV

<table>
<tr><th rowspan="3">Course Code</th><th rowspan="3">Course Title</th><th colspan="4">Teaching Scheme</th><th>Internal</th><th colspan="2">End Semester Examination</th><th rowspan="3">Total</th></tr>
<tr><th colspan="3">Contact Hours</th><th rowspan="2">Credit</th><th rowspan="2">Continuous Evaluation</th><th rowspan="2">Report</th><th rowspan="2">Presentation & Viva</th></tr>
<tr><th>Inst.</th><th>Industry</th><th>Total</th></tr>
<tr><td>CA941</td><td>Dissertation/ Project Work</td><td>2</td><td>28</td><td>30</td><td>30</td><td>200</td><td>200</td><td>400</td><td>800</td></tr>
</table>

DETAILED SYLLABUS

FOR

MCA PROGRAMME
(1st SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2022-23

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester - I

<table>
<tr><th rowspan="4">Course Code</th><th rowspan="4">Course Title</th><th colspan="4">Teaching Scheme</th><th colspan="7">Examination Scheme</th></tr>
<tr><th colspan="3">Contact Hours</th><th rowspan="3">Credit</th><th colspan="3">Theory</th><th colspan="3">Practical</th><th rowspan="3">Total</th></tr>
<tr><th rowspan="2">Theory</th><th rowspan="2">Practical</th><th rowspan="2">Total</th><th colspan="2">Internal</th><th rowspan="2">External</th><th colspan="2">Internal</th><th rowspan="2">External</th></tr>
<tr><th>Case Study</th><th>Tests</th><th>Term work</th><th>Tests</th></tr>
<tr><td>CA859-862</td><td>Elective-I</td><td>4</td><td>-</td><td>4</td><td>4</td><td>10</td><td>20</td><td>70</td><td>-</td><td>-</td><td>-</td><td>100</td></tr>
<tr><td>CA863</td><td>Enterprise Computing using Java EE</td><td>4</td><td>3</td><td>7</td><td>7</td><td>10</td><td>20</td><td>70</td><td>15</td><td>15</td><td>70</td><td>200</td></tr>
<tr><td>CA864</td><td>Programming with .NET Architecture</td><td>3</td><td>3</td><td>6</td><td>6</td><td>10</td><td>20</td><td>70</td><td>15</td><td>15</td><td>70</td><td>200</td></tr>
<tr><td>CA865</td><td>NoSQL Database</td><td>4</td><td>3</td><td>7</td><td>6</td><td>10</td><td>20</td><td>70</td><td>15</td><td>15</td><td>70</td><td>200</td></tr>
<tr><td>HS141.02 C</td><td>Foreign Languages</td><td rowspan="2">-</td><td rowspan="2">2</td><td rowspan="2">2</td><td rowspan="2">2</td><td colspan="2" rowspan="2">-</td><td rowspan="2">-</td><td colspan="2" rowspan="2">30</td><td rowspan="2">70</td><td rowspan="2">100</td></tr>
<tr><td>HS105.02 C</td><td>Academic Speaking and Presentation Skills</td></tr>
<tr><td colspan="2"></td><td>15</td><td>11</td><td>26</td><td>26</td><td colspan="3">400</td><td colspan="3">400</td><td>800</td></tr>
</table>

Elective-I	
Course Code	Course Title
CA859	Cloud Computing
CA860	Cryptography & Network Security
CA861	Block Chain Technology
CA862	Quantum Computing

CA859: Cloud Computing

(100 Marks)

Contact Hours: 04

Pre-requisite: Operating System Concepts and Network Technology

Methodology & Pedagogy: During theory lectures the emphasis will be given on the fundamentals of cloud computing. Students will be introduced basic types, architecture, service providers, mechanism, security issues and some hidden aspects of cloud computing. Students will give practical exposure in form of case study and by showing cloud infrastructure of university.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Evolution of Cloud Computing	06
2	Understanding Cloud Computing and basic types	09
3	Fundamentals of Cloud Architecture and Service Providers	09
4	Cloud Computing Mechanisms	09
5	Cloud Computing Security and Business Use	08
6	Intensive analysis of Cloud Computing with case studies	07

Total Hours (Theory): 48

Total Hours:48

Detail Syllabus:

Unit I : Evolution of Cloud Computing **Hours 06**

Introduction of Cloud Computing, Growth of technology, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

Unit II : Understanding Cloud Computing and basic types **Hours 09**

Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing. Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing.

Unit III : Fundamentals of Cloud Architecture and Service Providers **Hours 09**

Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture. Introduction to major Cloud Service Provider: Amazon Web Services, Google Cloud. Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terrimark, DropBox.

Unit IV : Cloud Computing Mechanisms **Hours 09**

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication. Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

Unit V : Cloud Computing Security and Business Use **Hours 08**

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single SignOn (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control. Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

Unit VI : Intensive analysis of Cloud Computing with case studies **Hours 07**

Overview of Cloud services, Designing Solutions for the Cloud, Implement & Integrate Solutions, Emerging Markets and the Cloud, Tools for Building Private Cloud: IaaS using Eucalyptus, PaaS on IaaS - AppScale

Core Books:

1. Kevin L. Jackson, Scott Goessling: Architecting Cloud Computing Solutions: Packt Publication: 2018
2. Thomas Erl, Zaigham Mahmood and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture, PHI, 2013.
3. S. Srinivasan: Cloud Computing Basics, Springer,2014.

Reference Books:

1. Derrick Rountree, Ileana Castrillo : The Basics of Cloud Computing, Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing- Principles and Paradigms, John Wiley &Sons, 2011.

Web References:

1. http://whatisCloud.com/basic_concepts_and_terminology/Cloud [For basic terminology of Cloud Computing]
2. http://www.tutorialspoint.com/Cloud_Computing/ [For cloud computing lecture notes]

3. http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloudcomputingvirtualization-building-private-iaas-guide.pdf [For cloud computing virtualization]
4. www.cs.purdue.edu/.../Anya-Kim-Bhargava-MCCWorkshop.ppt [Security issues PPTs]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Students will learn basics of cloud computing**
CO2 :	The students will be familiar with various cloud architectures and services.
CO3 :	They will get the knowledge of various network mechanics.
CO4 :	Students will get various business aspects of cloud computing with security aspects
CO5 :	They will be able to design and deploy Cloud Infrastructure

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	**Evolution of Cloud Computing**	√				
2	**Understanding Cloud Computing and basic types**		√			
3	**Fundamentals of Cloud Architecture and Service Providers**		√			
4	**Cloud Computing Mechanisms**			√		
5	**Cloud Computing Security and Business Use**				√	
6	**Hidden Aspects of Cloud Computing**					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	3	1	2	2	-	-	1	-
CO2	3	1	1	1	1	1	3	1	2	2	1	3	2	-
CO3	3	2	1	2	1	3	3	2	2	2	-	3	2	-
CO4	3	3	2	3	1	3	3	2	2	2	3	-	1	-
CO5	3	3	3	3	1	3	3	3	2	2	2	-	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA860: Cryptography & Network Security

(100 Marks)

Contact Hours: 04

Pre-requisite: Basic concepts of computer network and digital communication

Methodology & Pedagogy: During the lectures the students will learn about basic cryptographic concepts. The teacher will also present applications of cryptography with focus on network security. Further the teacher will also introduce system level security issues and their solutions.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to network security	08
2	Symmetric and public key cryptography systems and related concepts	10
3	Key distribution	09
4	Transport level security	09
5	Wireless and Internet security	06
6	Advanced Security Concepts	06

Total Hours (Theory): 48

Total Hours: 48

Detail Syllabus:

Unit I: Introduction to network security **Hours 08**

Network security concepts, OSI security architecture, security attacks, security services, security mechanism, network security standards.

Unit II: Symmetric and public key cryptography systems and related concepts **Hours 10**

Symmetric encryption system and algorithm, concepts of random and pseudo numbers, operations on stream ciphers and block ciphers. Message authentication concepts, secure hash function, message authentication codes, public key cryptography, and digital signature.

Unit III: Key distribution **Hours 09**

Basics of key distribution, symmetric key distribution, Kerberos algorithm, asymmetric key distribution, public key infrastructure, federated identity management.

Unit IV: Transport level security **Hours 09**

World Wide Web security, secure socket layer and transport layer security, Secure HTTP (HTTPS), Secure Shell (SSH).

Unit V: Wireless and Internet security **Hours 06**

Wireless security: IEEE 802.11 standard, wireless LAN security, wireless application protocol (WAP), transport layer security in wireless network, WAP end-to-end security.

Internet security: pretty-good-privacy, multipurpose internet mail system (MIME), domain key based mail identification. IP security policy, integrated security features in IP packet, internet key exchange (IKE), cryptographic suits.

Unit VI: Advanced security concepts **Hours 06**

Intrusion detection, password management, and malicious software: viruses, worms, need for firewall, characteristics of firewall, and types of firewall, firewall location and configuration.

Core Books:

1. William Stallings: Network Security Essentials: 4th Edition, Pearson Publication: 2018.
2. Bruce Schneier: Applied cryptography: 3rd Edition, Wiley Publication: 2016
3. William Stallings: Cryptography and network security: 6th Edition, Pearson Publication: 2019

Reference Books:

1. Alphred J. Menezes: Handbook of applied cryptography: 3rd Edition:CRC Press : 2010.
2. David Pointcheval: Applied cryptography and network security: 1st Edition: Springer: 2012
3. W. Chen, H.-W.LI et al.; Applied Cryptography and Network Security, 1st Edition, Magnum Publication, 2016.

Web References:

1. https://www.edureka.co/blog/what-is-network-security/ [Introduction to Network Security]
2. https://sectigostore.com/blog/5-differences-between-symmetric-vs-asymmetric-encryption/ [Symmetric and Asymmetric Key Difference]
3. https://www.cloudflare.com/en-in/learning/ssl/transport-layer-security-tls/ [Transport Level Security]
4. https://www.checkpoint.com/cyber-hub/network-security/what-is-firewall/ [Firewall]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Be able to understand Network security**
CO2 :	Be familiar with Symmetric and Public key cryptography and key distribution
CO3 :	Understand Transport level security
CO4 :	Be familiar with wireless and Internet security
CO5 :	Be able to understand Advanced security concepts

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	**Introduction to network security**	√				√
2	**Symmetric and public key cryptography systems and related concepts**		√	√		
3	**Key distribution**		√			
4	**Transport level security**			√	√	√
5	**Wireless and Internet security**	√		√		√
6	**Advanced security concepts**		√		√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	3	2	3	1	2	2	3	2	3	2	3	3
CO2	3	2	3	3	3	2	3	3	2	3	2	3	2	3
CO3	2	2	3	3	3	2	2	2	3	2	3	3	3	3
CO4	2	3	3	2	3	1	2	3	2	3	2	3	3	2
CO5	3	3	3	3	3	2	3	3	2	2	3	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA861: Blockchain Technology

(100 Marks)

Contact Hours: 04

Pre-requisite: None

Methodology & Pedagogy: During the lecture sessions, the students will learn about various sub systems that work in integrated manner to make blockchain work. The teacher will introduce students to concepts of bitcoin, ethereum and hyperledger technologies. The students will also understand about application domains where blockchain can be applied.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to blockchain	07
2	Functioning of blockchain	08
3	Bitcoin and related concepts	08
4	Ethereum and related concepts	09
5	Hyperledger and related concepts	09
6	Applications of blockchain	07

Total Hours (Theory): 48

Total Hours:48

Detail Syllabus:

Unit I : Introduction to blockchain — **Hours 07**

History of blockchain, Centralized, decentralized and distributed systems, Layers of blockchain, Importance of Blockchain

Unit II : Functioning of blockchain — **Hours 08**

Cryptographic concepts- Symmetric key cryptography, Cryptographic hash function, MAC and HMAC, Asymmetric key cryptography,Diffie-Hellman key exchange, Byzantine general's problem, Merkel tree, Structure of block

Unit III : Bitcoin and related concepts **Hours 08**

What is bitcoin, Working with bitcoin, Bitcoin network, Bitcoin script, Bitcoin mining, Bitcoin wallets

Types of blockchain networks, Consensus algorithms

Unit IV : Ethereum and related concepts **Hours 09**

Design of ethereum, Ethereum blockchain, Smart contracts in ethereum, Ethereum virtual machine, Ethereum eco system

Unit V : Hyperledger and related concepts **Hours 09**

Introduction to hyperledger and hyperledger fabric, hyperledger fabric transaction flow, membership and identity management, fabric network setup, hyperledger fabric demo, hyperledger composer demo.

Unit VI : Applications of blockchain **Hours 07**

Applications of blockchain in domains – Finance, Healthcare, Insurance, Supply chain management,Internet of things.

Core Books:

1. Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda: Beginning Blockchain - A Beginner's Guide to Building Blockchain Solutions: 1st Edition: APress Publication: 2018.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder: Bitcoin and Cryptocurrency Technologies - A Comprehensive Introduction: 1st Edition: Princeton University Press: 2016.
3. Tiana Laurence: Introduction to blockchain technology – the many faces of blockchain technology in the 21st century: 1st Edition: Van Heren Publishing: 2019

Reference Books:

1. S. Ashraf, S. Adarsh: Decentralized Computing Using Blockchain Technologies and Smart Contracts - merging Research and Opportunities: 1st Edition: IGI Global, 2017
2. Fabian Schar, Aleksander Berentsen: Bitcoin, Blockchain, and Cryptoassets: 1st Edition: MIT Press: 2020

Web References:

1. https://andersbrownworth.com/cms/460/blockchain/demo [Blockchain Demo]
2. https://ethdocs.org/en/latest/introduction/what-is-ethereum.html [Ethereum Introduction]
3. https://www.hyperledger.org/wp-content/uploads/2018/07/HL_Whitepaper_IntroductiontoHyperledger.pdf [Hypderledger Introduction]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**The students will learn about underlying technologies of blockchain and various applications of blockchain technology.**
CO2 :	The students will understand working of blockchain in detail.
CO3 :	The students will learn about how the blockchain is used to power bitcoin.
CO4 :	The students will see how ethereum utilizes blockchain.
CO5 :	The students will recognize the role of blockchain in hyperledger platform.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction to blockchain	√				
2	Functioning of blockchain		√			
3	Bitcoin and related concepts			√		
4	Ethereum and related concepts				√	
5	Hyperledger and related concepts					√
6	Applications of blockchain	√				

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	1	-	-	3	-	-	-	-	1	1	-
CO2	-	1	2	1	1	-	3	-	-	-	-	1	1	-
CO3	2	2	3	-	3	-	-	-	-	-	-	3	2	1
CO4	3	2	3	-	3	-	-	-	-	-	-	3	2	1
CO5	3	1	3	1	3	-	-	-	-	-	-	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA862: Quantum Computing

(100 Marks)

Contact Hours: 04

Pre-requisite: Basics of Physics, Mathematics with a focus on abstract linear algebra and basics of computer science knowledge

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of Quantum Computing and quantum mechanics with mathematics and physics. The Agile software development concepts will be introduced during lectures. As part of modeling with UML students will be given exposure to model real-life software projects.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Fundamentals of Quantum Computing	06
2	Introduction to Quantum Mechanics	09
3	Quantum Computation and Quantum Fourier Transform	09
4	Quantum Computers	09
5	Quantum Information	08
6	Quantum Error Correction	07

Total Hours (Theory):48

Total Hours:48

Detail Syllabus:

Unit I : Fundamentals of Quantum Computing **Hours 06**

Brief history about Quantum Computing with Turing Machines, Quantum bits, Bloch sphere representation of a qubit, multiple qubits. Global Perspectives, Quantum Bits, Introduction to Quantum Computation, Quantum Algorithms, Quantum Information, Postulates of Quantum Mechanisms.

Unit II : Introduction to Quantum Mechanics **Hours 09**

Linear algebra, tensor products, postulates of QM, The EPR Paradox and Bell's inequality theorem,

Reversible computation.

Unit III : Quantum Computation and Quantum Fourier Transform **Hours 09**

Quantum algorithms, Single qubits operations, controlled operations, States and measurements, Universal quantum gates, Quantum Fourier Transform and its application, phase estimation, order-finding and factoring. Quantum search algorithms – Quantum counting – Speeding up the solution of NP – complete problems – Quantum Search for an unstructured database.

Unit IV : Quantum Computers **Hours 09**

Guiding Principles, Conditions for Quantum Computation, Harmonic Oscillator Quantum Computer, Optical Photon Quantum Computer – Optical cavity Quantum electrodynamics, Ion traps, Nuclear Magnetic resonance.

Unit V : Quantum Information **Hours 08**

Quantum noise and Quantum Operations – Classical Noise and Markov Processes, Quantum Operations, Examples of Quantum noise and Quantum Operations – Applications of Quantum operations, Limitations of the Quantum operations formalism, Distance Measures for Quantum information.

Unit VI : Quantum Error Correction **Hours 07**

Introduction, Classical Error Correction, Theory of quantum error-correction, constructing quantum codes, Stabilizer Codes, Fault Tolerant Quantum Computation.

Core Books:

1. Micheal A. Nielsen. &Issac L. Chiang, “Quantum Computation and Quantum Information”, Cambridge University Press, Fint South Asian edition, 2002.
2. Eleanor G. Rieffel , Wolfgang H. Polak , “Quantum Computing - A Gentle Introduction” (Scientific and Engineering Computation) Paperback – Import, 3 Oct 2014
3. Computer Science: An Introduction by N. DavidMermin 5. Yanofsky's and Mannucci, Quantum Computing for Computer Scientists.
4. Nielsen M. A., Quantum Computation and Quantum Information, Cambridge University Press.

Reference Books:

1. Computer Science: An Introduction by N. DavidMermin
2. Yanofsky's and Mannucci, Quantum Computing for Computer Scientists

Web References:

1. https://www.springer.com/gp/campaigns/quantum-science-and-quantum-technology
2. https://link.springer.com/book/10.1007/978-1-4612-1390-1
3. https://www.worldscientific.com/doi/pdf/10.1142/9789812385253_fmatter
4. Kitaev A.Y., Shen A.H., Vyalyi M.N. Classical and quantum computation (GSM047, AMS, 2002)(ISBN 9780821832295)(O)(274s) PQm .pdf [EBook]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Distinguish problems of different computational complexity describe the architecture of a quantum system.**
CO2 :	comprehend the principles of mathematics and physics of quantum computation
CO3 :	identify applications of quantum computing
CO4 :	Produce documentation that is comprehensible to a group of different programmers and present the theoretical background
CO5 :	apply various security measures for quantum communication and error detection and correction

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	**Fundamentals of Quantum Computing**	√				
2	**Introduction to Quantum Mechanics**		√			
3	**Quantum Computation and Quantum Fourier Transform**			√		
4	**Quantum Computers**			√		
5	**Quantum Information**				√	
6	**Quantum Error Correction**					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	3	-	-	-	-	-	-	-	-	-
CO2	3	-	-	1	-	2	-	-	-	-	-	-	-	2
CO3	2	3	2	-	1	1	3	-	3	-	-	-	3	-
CO4	-	3	3	-	2	-	-	-	-	-	-	2	2	-
CO5	-	2	3	3	1	-	2	2	1	1	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA863: Enterprise Computing using Java EE

(200 Marks)

Contact Hours: 07

Pre-requisite: Fundamental Knowledge of Core Java (Java SE) concepts.

Methodology & Pedagogy: Advanced Java fundamentals like JDBC, Servlet, JSP and Hibernate provides capabilities and features for developing web application which is essential for developer in order to understand entire web application development life cycle. During the theory sessions of the syllabus, fundamentals of the mentioned technologies will be explained in detail with real time examples. Moreover, practical session will provide hands-on exposure about JDBC, Servlet, JSP and Hibernate with variety of practical assignment. Also, MVC architecture using these technologies will be covered during theory as well as practical sessions.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	JDBC Programming	06	36
2	Web Application Components Fundamentals	05	
3	Servlet API	08	
4	Java Server Pages	05	
5	Basics of Hibernate 4.0	07	
6	Packaging and Deploying Web Applications	05	

Total Hours (Theory): 36

Total Hours (Lab) : 36

Total Hours: 72

Detail Syllabus:

Unit I : JDBC Programming **Hours 06**

The JDBC Connectivity Model, Database Programming: Connecting to the Database, creating a SQL Query, Getting the Results, Updating Database Data, Error Checking and the SQLException Class, the SQLWarning class, the Statement Interface, PreparedStatement, CallableStatement, the ResultSet Interface, Updatable Result sets, JDBC Types, Executing SQL Queries, ResultSetMetaData, Executing SQL Updates, Transaction Management

Unit II : Web Application Components Fundamentals **Hours 05**

Basics of web application development and web components, Introduction of MVC design pattern, Fundamentals of: Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, HTTP request methods, HTML Form Processing, HTTP request response cycle.

Unit III : Servlet API **Hours 08**

Servlet Model: Overview of Servlet, Servlet Life Cycle, HTTP Methods Structure and Deployment descriptor ServletContext and ServletConfig interface, Attributes in Servelt, Request Dispacher interface

The Filter API: Filter, FilterChain and Filter Config

Cookies and Session Management: Understanding state and session, Understanding Session Timeout and Session Tracking, URL Rewriting

Unit IV : Java Server Pages **Hours 05**

JSP Overview: The Problem with Servlets, Life Cycle of JSP Page, JSP Processing, JSP Application Design with MVC, Setting Up the JSP Environment

JSP Directives, JSP Action, JSP Implicit Objects JSP Form Processing, JSP Session and Cookies Handling, JSP Session Tracking

JSP Database Access, JSP Standard Tag Libraries, JSP Custom Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing

Unit V : Basics of Hibernate 4.0 **Hours 07**

Overview of Hibernate, Hibernate Architecture, Hibernate Mapping Types, Hibernate O/R Mapping, Hibernate Annotation, Hibernate Query Language

Unit VI : Packaging and Deploying Web Applications **Hours 05**

Basics of server container, build a WAR artifact for the web application, configure the Docker connection node and deploy the application to server container

Core Books:

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.
2. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.
3. Jeff Linwood and Dave Minter: Hibernate 2nd edition Beginning Après publication

Reference Books:

1. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition
2. James Keogh: Complete Reference J2EE, Mcgraw publication

Web References:

1. http://courses.coreservlets.com/Course-Materials/csajsp2.html [Servlet Basics]
2. http://www.ceit.es/asignaturas/InteInfo/Recursos/Servlets/JavaServlets.pdf [Servlet Tutorial PDF]
3. http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf [JDBC Tutorial]
4. www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt [JSP Tutorial Slides]

Course Outcomes: Upon successful completion of the course, students will:

CO1 :	**Able to connect Java application with various RDBMS using JDBC**
CO2 :	Have understanding of web application architecture and terminologies
CO3 :	Able to develop real time web application using Servlet and JSP technologies
CO4 :	Able to use Object Relation Mapping using Hibernate to build database
CO5 :	Apply Model-View-Controller architecture to build complex client-server applications

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	JDBC Programming	√		√		
2	Web Application Components Fundamentals		√	√		
3	Servlet API			√	√	√
4	Advanced Servlet Features and Security			√	√	√
5	Java Server Pages			√	√	√
6	Packaging and Deploying Web Applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	1	1	-	-	-	-	-	-	-	3	2
CO2	3	2	3	2	-	-	-	-	-	-	2	1	2	2
CO3	3	2	3	2	2	-	-	-	-	2	2	3	3	3
CO4	3	2	3	2	2	2	-	-	1	-	2	1	3	2
CO5	2	2	3	3	2	-	1	1	-	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA864: Programming with .NET Architecture

(200 Marks)

Contact Hours: 06

Pre-requisite: Familiarity with basic concepts of object oriented programming

Methodology & Pedagogy:

In the lectures students will learn about basics of .NET framework, ASP.NET and other advanced concepts related to programming with .NET framework. In the lab sessions the student will learn about solving real world problems with .NET framework and related technologies.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to .NET framework and ASP.NET	**04**	**36**
2	Developing applications with ASP.NET	**07**	
3	Database connectivity in ASP.NET	**08**	
4	MVC architecture in .NET framework	**06**	
5	SOA in .NET framework	**05**	
6	Advanced concepts in ASP.NET	**06**	

Total Hours (Theory): 36

Total Hours (Lab):36

Total Hours: 72

Detail Syllabus:

Unit I : Introduction to .NET framework and ASP.NET **Hours 04**

Introduction to .NET framework, Introduction to Web Programming, Introduction to ASP and ASP.NET, Deploying ASP.NET Application

Unit II : Developing applications with ASP.NET **Hours 07**

ASP.NET Page Life Cycle, Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Developing a web form, Working with master pages, State management techniques, Application Tracing, Error Handling

Unit III : Database connectivity in ASP.NET **Hours 08**

Introduction to ADO and ADO.NET, ADO.NET architecture, Connection Oriented and Connection Less architecture, Binding data to web controls and working with data controls, working with XML

Unit IV : MVC architecture in .NET framework **Hours 06**

Introduction to MVC Architecture in ASP.NET, CRUD operation using MVC architecture, working with web controls in MVC architecture, introduction to LINQ.

Unit V : SOA in .NET framework **Hours 05**

Introduction to service oriented architecture concept, SOAP protocol, Concept of UDDI and WSDL, Applications of SOA, Building SOA application with ASP.NET.

Unit VI : Advanced concepts in ASP.NET **Hours 06**

Introduction to ASP.NET AJAX, AJAX control toolkit and extender controls. Introduction to .NET core, characteristics of .NET core, difference between .NET and .NET core.

Core Books:

1. Cogent Solutions Inc.: ASP.Net 4.5 Black book, Dreamtech press, 2019
2. Mridila Parihar, Essam Ahmed: ASP .Net Bible, Wiley, 2017
3. Stephon Walther: ASP.Net Unleashed, BPB publication, 2018

Reference Books:

1. Mesbah Ahmed, Chris Garrett, Jeremy Faircloth, Chris Payne: ASP.Net Programming. Developer's Guide, Dreamtech, 2016.
2. Greg Buczek: ASP.Net Tips & Techniques, Tata McGraw Hill Edition, 2016.
3. Bolton, Justin Langford, Glenn Berry, Gavin Payne, Amit Banerjee, Rob Farley: Professional SQL Server 2019 internals and troubleshooting, Wiley India publication, October, 2019.

Web References:

1. http://msdn.microsoft.com/en-us/aa336522.aspx [For software download]
2. https://docs.microsoft.com/en-us/aspnet/overview [For ASP.NET technical documentation]
3. https://www.tutorialspoint.com/dotnet_core/dotnet_core_overview.htm [for .NET Core]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Students will be able to understand .NET framework, ASP.NET technologies and develop simple applications using the framework.**
CO2 :	Students will be able to develop database centric applications.
CO3 :	Students will be familiar with concepts of MVC for application development.
CO4 :	Students will able to understand SOA concepts and develop simple SOA based applications with .NET framework.
CO5 :	Students will be familiar with the concepts of AJAX and other advance .net technology concepts.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to .NET framework and ASP.NET	√				
2	Developing applications with ASP.NET	√				
3	Database connectivity in ASP.NET		√			
4	MVC architecture in .NET framework			√		
5	SOA in .NET framework				√	
6	Advanced concepts in ASP.NET					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	3	-	2	1	-	-	2	-	-	2	1
CO2	2	3	2	2	-	2	1	-	-	2	2	1	3	1
CO3	1	2	2	-	-	-	-	-	-	-	2	1	3	2
CO4	2	3	3	-	2	-	-	-	-	-	2	1	3	2
CO5	3	2	3	2	-	-	3	-	-	-	-	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA865: NoSQL Database

(200 Marks)

Contact Hours: 07

Pre-requisite: Basics of Database Management Systems and Data Structure

Methodology & Pedagogy: During theory lectures student will understand the essential concepts that act as the building blocks for many of the NoSQL products. It starts from the fundamentals of NoSQL and graduates to advanced concepts of architecture, storage internals and indexing. During Practical sessions, students will be required to design the applications using NoSQL Databases.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to NoSQL	09	36
2	NoSQL Stores Data Models	08	
3	NoSQL Storage Architecture	06	
4	Working with Data using MongoDB	08	
5	Querying CRUD operation using MongoDB	09	
6	Database Administration using MongoDB	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to NoSQL **Hours 09**

Introduction to SQL Relational Model, Codd Rules, Overview and History of NoSQL Databases, Taxonomies by Data Model, Overview of No-SQL Storage : Key Value, Document-Oriented, Column-Family and Graph-storage. Techniques and Pattern: CAP Theorem, ACID Vs. BASE properties and Comparison of relational databases to new NoSQL stores.

Unit II : NoSQL Stores Data Models **Hours 08**

MongoDB, Cassandra, HBASE, Neo4j use and deployment, Application, Challenges NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Aggregate-Oriented Databases. Replication and Sharding, Overview of MapReduce on databases. Distribution Models, Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication

Unit –III NOSQL- Storage Architecture **Hours 06**

Working with Column-Oriented Databases, Using Tables and Columns in Relational Databases, Contrasting Column Databases with RDBMS, Column Databases as Nested Maps of Key/Value Pairs, HBase Distributed Storage Architecture, Document Store Internals, Storing Data in Memory-Mapped Files, Guidelines for Using Collections and Indexes in MongoDB, MongoDB Reliability and Durability, Horizontal Scaling, Understanding Key/Value Stores in Redis.

Unit IV: Working with Data using MongoDB **Hours 08**

Overview of MongoDB, Document Databases, What Is a Document Database? Installing MongoDB on Your System, Running MongoDB, Installing Additional Drivers, Introduction to MongoDB key features, Core Server tools, MongoDB through the JavaScript's Shell and others.

Unit V: Querying CRUD operation using Mongodb **Hours 09**

Navigating your databases, viewing available data and collections, inserting data, querying data, using sort, limit and skip functions, retrieving single document, using aggregation commands, working with condition, updating data, upsert and save command, rename a collection, referencing a database

Unit VI : Database Administration using MongoDB **Hours 08**

MongoDB server, backup a single database, single collection, restoring the collection, importing data into mongo- dB, exporting data from mongodb, securing your data, restricting access, protecting your server with authentication.

Core Books:

1. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Author: Sadalage, P. & Fowler, Publication: Pearson Education.

Reference Books:

1. Definite Guide to MongoDB by Eelco Plugge, Peter Membrey and Tim Hawkins, Apress.
2. Getting started with NOSQL by Gaurav Vaish, PACKT.
3. Name: Redmond, E. &Wilson, Author: Seven Databases in Seven Weeks: A Guide to Modern Databases and the NoSQL Movement Edition: 1st Edition.
4. NOSQL theory and Examples by Piotr fulmanski, Simple introduction series: e-book.
5. Guy Harrison, Next Generation Database: NoSQL and big data, Apress.

Web References:

1. https://www.tutorialspoint.com/mongodb_for_absolute_beginners/index.asp [MongoDB For Absolute Beginners - Tutorialspoint]
2. https://www.tutorialspoint.com/neo4j/neo4j_data_model.htm [Neo4j - Data Model (tutorialspoint.com)]
3. https://appinventiv.com/blog/hbase-vs-cassandra/[HBase vs Cassandra: Which is The Best NoSQL Database (appinventiv.com)]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Able to understand the need and what is NoSQL**
CO2 :	Able to define, compare and use the four types of NoSQL Databases
CO3 :	Able to demonstrate an understanding of the detailed architecture, define objects, load the data, and query data
CO4 :	Able to design Schema and implement CRUD operations, distributed data operations, and implement various column store internals
CO5 :	Able to develop Application with Graph Data model

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction to NoSQL	√				
2	NoSQL Stores with Replication & Sharding		√			
3	NoSQL Key-Value Database using MongoDB			√		
4	NoSQL Key-Value Database using Riak			√		
5	NoSQL Columnar Database using HBASE & Cassandra				√	
6	NoSQL Graph Modeling using Neo4j					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	1	-	-	-	-	-	-	-	-	-	-
CO2	3	-	2	2	-	-	-	3	2	-	-	-	1	2
CO3	3	3	3	-	1	-	-	-	3	-	-	-	3	-
CO4	3	-	2	-	2	-	2	2	2	-	3	1	1	1
CO5	3	3	1	3	3	-	-	3	-	-	1	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS141.02 C: FOREIGN LANGUAGES (French)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/01	--	30/15	**02**
Marks	**--**	100	--	100	

Pre-requisite courses:

- French Language Studies- Introduction (Coursera)

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to French Language	08
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Introduction to French Language	08 Hours	28%
	Facts and figures about French Language; Basic French Linguistics-* Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English; **Grammar-**Subject Pronoun, Verbs: (être, avoir, habiter, regarder, manger … "er" verb), Form of address, Numbers (1 to 20), Nouns and plurals of nouns, The expression: C'est, Il y a; **Presentation:** -1) Self-Introduction-2) Question and answering; **Dialogue**		
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08 Hours	28%

	Grammar-Definite articles**,** Indefinite articles**,** Present tense (Positive Forms**,** Negative Forms)**,** Numbers (21 to 100, 100-1000)**,** Days, Months, Family, Verbs: (aller, venir, finir, pouvoir, vouloir … “ir” verb)**; Social Links-**1), My family & relations 2) Appointments 3) Gathering information from someone**; Dialogue**		
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08 Hours	28%
	Grammar- Common Adjectives**,** Comparative Adjectives**,** Common Adverbs**,** Interrogative Forms**,** The expression: “On”**,** Directions, Countries, Nationalities, Seasons, Weather, Professions**,** Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “ verb)**; Work , Study and Travel-**1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport)**; Dialogue**		
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06 Hours	26%
	Grammar-1) Common Prepositions 2) Common Conjunctions 3)Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: (“er”, “ir”,”re” etc...)**; Food & Shopping-**1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/At Restaurant / At Market)**; Dialogue**		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Gain basic communication skills in French language with preliminary understanding of grammar
CO2	Develop vocabulary required to speak about him/herself and his/her immediate environment.
CO3	Become capable of interacting in simple ways, to ask simple questions to get necessary information, to reply simple questions.
CO4	Become capable of understanding and using simple instructions in their personal, academic and professional environments.
CO5	Develop skills and intelligences to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new global trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- **Text book:**
  1. Complete French: All-In-One, McGraw-Hill, Amazon
  2. Best for Grammar: Easy French Step-by-Step, McGraw-Hill, Amazon
- **Reference book:**
  1. Basic French: McGraw-Hill, Amazon
  2. French Grammar for Beginners, Amazon
- **Web material:**
  1. https://alison.com/course/french-language-studies-introduction
  2. https://alison.com/course/basic-french-language-skills-for-everyday-life-revised-2017
  3. http://www.bbc.co.uk/languages/french/
  4. https://www.loecsen.com/en/learn-french
  5. https://www.youtube.com/watch?v=ujDtm0hZyII

HS105.02 C: ACADEMIC SPEAKING AND PRESENTATION SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	**02**
Marks	**--**	100	--	100	

Pre-requisite courses:

- Beginner/Intermediate level language proficiency

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Foundations of Advance Communication	04
2.	Art of Conversation	06
3.	Science of Power Speaking	06
4.	Academic Speaking Application – Part I	08
5.	Academic Speaking Application – Part II	06
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Foundations of Advance Communication	04 Hours	14%
	Meaning and Definition of Advance Communication; Advance Communication in Digital, Social, Mobile World; Strategies for Advance Communication; Meaning and Concept of Academic Language; High Frequency Academic Vocabulary		
2.	Art of Conversation	06 Hours	20%
	Describing people, places and things; Expressing opinions; Making suggestions; Persuading someone; Interpreting and Summarizing		
3.	Science of Power Speaking	06 Hours	20%
	Phonemes, Word Stress, Pronunciation, Intonation, Pause, Register, Fluency, Prosody, Lexical Range		
4.	Academic Speaking Application – Part I	08 Hours	26%
	Art of Oratory, Formal Presentation, Speech Analysis – Decoding Best Speeches		
5.	Academic Speaking Application – Part II	06 Hours	20%
	Job Interview, Group Discussion, Meeting		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	understand and demonstrate advance communication and academic speaking skills
CO2	demonstrate ability to communicate in diverse situations
CO3	activate and extend their linguistic and communicative competence
CO4	demonstrate the formal presentation skills
CO5	demonstrate performing ability at group discussion and personal interview

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	2	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	3	-	-	-

Recommended Study Material:

- **Text book: -**
- **Reference book:**
  1. *Business Communication Today* (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
  2. *Effective Speaking Skills* by Terry O' Brien
  3. *Speak Better Write Better* by Norman Lewis
  4. *Well Spoken: Teaching Speaking to All Students* by Erik Palmer
  5. *Let Us Hear Them Speak : Developing Speaking – Listening Skills in English* by Jayshree Mohanraj (Publisher – Sage Publication)
  6. *The craft of scientific presentations: Critical steps to succeed and critical errors to avoid*. New York: Springer by Michael Alley
  7. *Presentation Skills in English* by Bob Dignen (Publisher: Orient Black Swan)
- **Web material:**
  1. TED Talk : How to speak so that people want to listen https://www.ted.com/talks/julian_treasure_how_to_speak_so_that_people_want_to_listen?language=en
  2. TED Talk: The 110 techniques of communication and public speaking

https://www.ted.com/talks/david_jp_phillips_the_110_techniques_of_communication_and_public_speaking

DETAILED SYLLABUS

FOR

MCA PROGRAMME
(2nd SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2022-23

Teaching and Examination Scheme (MCA Programme)
Effective from Year 2022-23
Semester-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA866-869	Elective-II	3	3	6	6	10	20	70	15	15	70	200
CA870	Software Quality Assurance	3	3	6	6	10	20	70	15	15	70	200
CA871	Advanced Mobile Programming	3	3	6	6	10	20	70	15	15	70	200
CA872	Software Engineering with Agile and DevOps	4	-	4	4	10	20	70	-	-	-	100
HS106.02 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-III **	-	2	2	2	-		-	30		70	100
		13	13	26	26	400			500			900

Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Internet and Web Designing and Mobile Application Development** course for others.

Elective-II	
Course Code	Course Title
CA866	Game Development using Unity
CA867	Python Web framework
CA868	Framework and Applications
CA869	HTTP Web Service for Enterprise Application

No	Course Code	Course Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	MB651	Software based Statistical Analysis	Management
5	NR755	First Aid & Life Support	Nursing
6	OC733.01	Introduction to Polymer Science	Applied Science
7	MA771.01	Reliability and Risk Analysis	Mathematics
8	ME781.01	Occupational Health &Safety	Engineering
9	MA772.01	Design of Experiments	Mathematics
10	RD701.01	Introduction to Analytical Techniques	Applied Science
11	RD702.01	Introduction to Nanoscience And Technology	Applied Science
12	PH891	Community Pharmacy Ownership	Pharmacy
13	PH892	Intellectual Property Rights	Pharmacy
14	PSE55	Astrophysics ,Space and Cosmos	Applied Science

CA866: Game Development using Unity

(200 Marks)

Contact Hours: 06

Pre-requisite: Basics of Animation, C# Language, Mathematics and Physics concepts

Methodology & Pedagogy: During theory lectures illustrations Graphics, animation and various concepts regarding Game Development. Emphasize will be given on some mathematical and physics concepts, Fundamental of Graphics and objects creation, 3D graphics, Collision detections, Fundamental of Game programming, Game loop, Game Engine and many more. During Practical sessions, students will develop 3D games using Unity.

Outline of the Course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Basics of Game Development in Unity	07	36
2	Game Development Tools and Resources	07	
3	Game Scene Designing	06	
4	Game Interface Designing	05	
5	Game Rendering and Walkthrough	06	
6	Game Testing and Deployment	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Basics of Game Development in Unity **Hours 07**

Getting started with 3D, Coordinates, Local space versus World space, Vectors, Cameras, Polygons, edges, vertices, and meshes, Materials, textures, Shaders, Rigid Body physics, Collision detection. Essential Unity concepts, Introduction to Assets, Scenes, Game Objects, Components, Scripts, Prefabs. The Interface: Scene window and Hierarchy, Inspector, Project and Game window.

Unit II: Game Development Tools and Resources **Hours 07**

The terrain editor: menu features, importing and exporting height maps, Terrain Toolset: Raise, plain and smooth the height, Paint texture, place trees, and terrain settings, creating island: use of Sun, Sea and Sand. Importing Model package. Player characters: working with inspector, Tags, Layers, Prefabs and the Inspector, Deconstructing the First Person Controller object, Parent-child issues, First Person Controller objects. Scripting basics: Variables, Functions, If Else, comments and major syntax. The FPS walker script

Unit III: Game Scene Designing **Hours 06**

Discovering Collisions, ray casting, Adding the outpost, Opening the outpost, Collision detection and creating new assets, attaching a script. Ray casting with disabling collision detection. Prefab. Collection and HUD: Creating the battery prefab. Download, import, and place, Tagging the battery, Scale, collider, and rotation, scattering batteries, Displaying the battery GUI, Creating the GUI Texture object, Positioning the GUI Texture, Scripting for GUI change, Battery collection with triggers, Restricting outpost access, Restricting access Hints for the player, Creating the fire particle systems

Unit IV: Game Interface Designing **Hours 05**

Interfaces and menus, Making the main menu, Adding the play button, Disabling Game Objects, Writing an OnGUI() script for a simple menu, Flexible positioning for GUIs, Adding UnityGUI buttons, Opening scenes with custom functions, GUI skin settings.

Decision time

Unit V: Game Rendering and Walkthrough **Hours 06**

Downloading assets, Making the smoke material, Particle system settings, Ellipsoid Particle Emitter settings, Particle Animator settings, Adding audio to the volcano, Volcano testing, Coconut trails, Editing the Prefab, Trail Renderer component, Updating the prefab, Performance tweaks, Camera Clip Planes and fog, Ambient lighting Instructions scene, Adding screen text, Text Animation using Linear Interpolation (Lerp), Menu return, Island level fade-in, UnityGUI texture rendering, Game win notification.

Unit VI: Game Testing and Deployment **Hours 05**

Build Settings, Web Player, Player Settings, Web Player Streamed, OS X Dashboard Widget, OS X/Windows, Standalone, Building the game, adapting for web build, Texture compression and debug stripping, building standalone 266Indie versus Pro, building for the Web, adapting web player builds, Quality Settings, Player Input settings, sharing your work, Testing and finalizing: Public testing

Core Books:

1. Will Goldstone: Unity Game Development Essentials: Edition 2nd: Packt Publication: 2009.
2. Greg Lukosek, “Learning C# by Developing Games with Unity 5.x”, 2nd Edition, Packt Publishing, 2016
3. Ashley Godbold, Simon Jackson, “Mastering Unity 2D Game Development”, 2nd Edition, Packt Publishing, 2016.
4. Joe Hocking, “Unity in Action: Multiplatform game development in C#”, 2nd Edition, Manning Publications, 2018
5. Sue Blackman, “Beginning 3D Game Development with Unity 4: All-in-one, multi-platform game development”, 2nd Edition, Apress, 2013

Reference Books:

1. Nicolas Alejandro Borromeo: Hands-On Unity 2021, Game Development: Edition 2nd: Packt Publication: 2021.
2. Ben Tristem, Mike Geig, “Unity Game Development in 24 Hours- Sams Teach Yourself” ,2nd Edition, Paperback, December 19, 2015
3. Michelle Menard, “Game Development with Unity”, Course Technology, 2013.
4. Matt Smith, Chico Queiroz, “Unity 5.x Cookbook”, Packt Publishing, October 5, 2015
5. Francesco Sapio, “Unity UI Cookbook”, Packt Publishing, 2015
6. P Patrick Felicia, “Unity 5 from Zero to Proficiency: A step-by-step guide to creating your first game", CreateSpace Independent Publishing Platform, February 25, 2016
7. Matt Smith, “Unity 2018 Cookbook”, 3rd Edition, Packt Publishing, August 31, 2018
8. Alan Thorn, “Mastering Unity Scripting”, Packt Publishing, January 29, 2015

Web References:

1. https://subscription.packtpub.com/search?query=unity
2. https://itsourcecode.com/free-projects/python-projects/mario-game-in-python-with-source-code/
3. http://learn.unity.com
4. https://unity3d.com/learn/tutorials/topics/developer-advice/how-start-your-game-development [Game Tutorial]
5. https://www.studytonight.com/game-development-in-2D/ [Game Tutorial]
6. https://msdn.microsoft.com/en-us/magazine/dn759441.aspx [Game Tutorial]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**able to gain a basic understanding of game development using Unity 3D**
CO2 :	able to learn concepts such as manipulating objects, scripting, and compiling
CO3 :	able to learn graphics and visuals in game development
CO4 :	able to develop script writing for any problem and solution as game
CO5 :	able to develop full 3D game

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Basics of Game Development in Unity	√				
2	Game Development Tools and Resources		√			
3	Game Scene Designing		√			
4	Game Interface Designing			√		
5	Game Rendering and Walkthrough				√	
6	Game Testing and Deployment					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	1	2	-	2	1	1	-	1	1	2	3
CO2	3	2	-	2	2	-	2	1	1	-	1	1	2	1
CO3	3	1	1	3	2	-	2	1	1	-	2	2	2	1
CO4	3	2	2	2	2	1	2	3	1	2	3	3	3	2
CO5	3	3	3	2	2	1	2	3	1	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA867: Python Web framework

(200 Marks)

Contact Hours: 06

Pre-requisite: Basics of Python Programming.

Methodology & Pedagogy: Theory sessions are required to address computational power of python through framework like DJango. The application of the topics taught in theory sessions is demonstrated in practical sessions. The case study will assist students in developing one working module in any paradigm using Python. The practical session should include actual application examples as well as MVT pattern integration.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Django	5	36
2	Django URL Patterns and Views	7	
3	Model and Database Setup for Django App	7	
4	Django Forms, Django & REST APIs	6	
5	Unit Testing with Django	6	
6	Admin, Security, Performance & Optimization	5	

Total Hours (Theory): 36

Total Hours (Lab) :36

Total Hours:72

Detail Syllabus:

Unit I: Introduction to Django **Hours 05**

Overview of Django, Python in context of Django, Features of Django, Model, View and Template pattern for Web Programming using Django, Installation guide for Django, First Application in Django

Unit II: Django URL Patterns and Views **Hours 07**

Designing a good URL scheme. URLconfTricks, Streamlining Function, Imports, Using Multiple View Prefixes, Special-Casing URLs in Debug Mode, Using Named Groups, Understanding the Matching/Grouping Algorithm, Passing Extra Options to View Functions Using Default View Arguments, Special-Casing Views, Capturing Text in URLs, Determining What the URLconf Searches Against, Including Other URLconfs, How Captured Parameters Work with include (), How Extra URLconf Options Work with include().

Unit III: Model and Database Setup for Django App **Hours 07**

Defining Model using ORM approach, Relationship between Models, Model Inheritance, Meta inner model, Setting up database. Django with PostgreSQL, SQLite3, MySQL. Django without Database. Django Project with Development Server, Creating and Updating database using manage.py

Unit IV: Django Forms, Django & REST APIs **Hours 06**

Form classes, Validation, Authentication, Advanced Forms processing techniques: Search forms, Creating Feedback form, Submission process of the form, Custom validation rules, Custom look and feel, Creating forms for models. REST APIs and its implementation. Django REST framework

Unit V: Unit Testing with Django **Hours 06**

Overview / Refresher on Unit Testing and why it’s good, Using Python’s unittest2 library, Test, Test Databases, Doctests, Debugging Best Practices

Unit VI: Admin, Security, Performance & Optimization **Hours 05**

Admin characterization in Django, Security facility in Django, Performance issues Django and Optimization in Django framework.

Core Books:

1. Adrian Holovaty, Jacob K. Moss: The Definitive Guide to Django:Web Development Done Right
2. Jeff Forcier, Paul Bissex, Wesley Chun: Python Web Development with Django, Pearson Education,2008.
3. Dusty Philips: Python 3 Object oriented Programming , PACKT publishing, 2010.

Reference Books:

1. Adrian Holovaty, Jacob Kaplan-Moss, et al.: The Django Book, 2013
2. Nathan Yergler: Effective Django, 2015
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. https://www.djangoproject.com/start/ [For Django Projects]
2. https://djangoforbeginners.com/initial-setup/ [For Django Setups]
3. https://docs.djangoproject.com/en/2.1/intro/tutorial01/ [For Django app]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Understand Django fundamentals and use its concepts to build and deploy robust web applications and apps.**
CO2 :	Learn about Django URL patterns and views and deploy Django applications.

CO3 :	Learn how to configure Django for powerful databases and create the Django admin interface.
CO4 :	Learn about Django’s security implications and how to create safe web applications with it.
CO5 :	To get some idea about unit testing concepts and its implementation through Django.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	**Introduction to Django**	√				
2	**Django URL Patterns and Views**		√			
3	**Model and Database Setup for Django App**			√		
4	**Django Forms, Django & REST APIs**				√	
5	**Unit Testing with Django**				√	
6	**Admin, Security, Performance & Optimization**					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	2	3	2	1	1	1	1	3	2
CO2	2	3	3	3	3	2	3	2	1	1	1	1	3	2
CO3	2	3	3	3	3	2	3	3	2	1	2	1	3	2
CO4	3	3	3	3	3	3	3	3	2	2	3	2	3	2
CO5	3	3	3	3	3	3	3	3	1	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA868: Frameworks and Applications

(200 Marks)

Contact Hours: 06

Pre-requisite: Java Enterprise Computing.

Methodology & Pedagogy: During theory sessions the students shall be introduced to various frameworks. Details of Spring and Hibernate frameworks will be discussed and their integration to develop real world applications will be demonstrated. During practical sessions students will be trained to develop various standalone and web applications using the studied frameworks.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Spring	04	36
2	Beans and Containers	07	
3	Spring Annotations	05	
4	Aspect-Oriented Programming	06	
5	Spring and Persistence	07	
6	Spring Web MVC and Spring Boot	07	

Total Hours (Theory): 36

Total Hours (Lab) : 36

Total Hours: 72

Detail Syllabus:

Unit I : Introduction to Spring **Hours 04**

Introduction of Framework, Characteristics of framework, Types of framework (Existing frameworks), Spring introduction, architecture of spring framework, spring framework modules overview, configuring spring framework, introduction of factory pattern, comparison between normal Java application and spring framework application.

Unit II : Beans and Containers **Hours 07**

What is inversion of control? Spring inversion of control overview, Spring Containers, Spring Configuration File, Spring Beans, Using the Container, The BeanFactory Interface, Singleton vs. Prototype, Bean Naming, Dependency Injection, Setter Injection, Constructor Injection

Unit III : Spring Annotations **Hours 05**

Annotations overview, default component names, inversion of control with annotations, setter and constructor injections with annotations, method and field injections with annotations, bean scope annotations, spring configuration with java code.

Unit - IV: Aspect-Oriented Programming **Hours: 06**

AOP Concepts, Join Points, Point Cuts, Advice, Configuration of Aspects - Types of Advice,AOPExample.

Unit V : Spring and Persistence **Hours 07**

Working with the HSQLDB Database, Integration with JDBC, Use of JdbcTemplate Class, Exception Translation, Updating with the JdbcTemplate Queries using the JdbcTemplate, Mapping Results to Java Objects.

Introduction to Object Relational Mapping, What is Hibernate?, The HibernateTemplate class, Hibernate Configuration Files, Mapping Classes and Fields for Hibernate, Creating and Saving a New Entity, Locating an Existing Entity, Updating an Existing Entity, Hibernate Sessions, Hibernate Query Language, Executing Queries, Hibernate annotation.

Unit VI : Spring MVC and Spring Boot **Hours 07**

What is Spring Web MVC?, Setting Dispatchers, Loading Configuration Files, Writing a Controller,

Types of Controller, Configuring the Controller, Setting of Handler Mapping, Handler Mapping Options, Handling a Form

Integrating Hibernate with Spring MVC – Accessing Database, Storing Form Values and Retrieving Data from Database.

Introduction to Spring Boot - Spring Boot Features, Spring Boot Application.

Core Books:

1. Craig Walls, Ryan Breidnbach: Spring in Action**,** 3rd Edition.
2. Rod Johnson, Juergen Hoeller, Alef Arendsen, Thomas Risberg, Colin Sampaleanu: Professional Java Development with the Spring Framework.

Reference Books:

1. Rod Johnson: J2EE Applications Without EJB**,** Wiley Publication.
2. API Documentation (http://www.springsource.org/spring-framework#documentation).

Web References:

1. https://www.udemy.com/course/spring-hibernate-tutorial/

2. http://netbeans.org/kb/docs/web/quickstart-webapps-spring.html[Spring Web MVC]
3. https://www.javatpoint.com/spring-boot-tutorial [Introduction to Spring Boot].

Course Outcomes: Upon successful completion of the course, students will:

CO1 :	**Be familiar with basics of Frameworks and advantages of frameworks**
CO2 :	Be able to work with Dependency Injection and IOC
CO3 :	Be able to use ApplicationContext to achieve DI.
CO4 :	Be familiar with AOP fundamental.
CO5 :	Be able to integrate NoSQL database with MVC based web application

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Spring	✓				
2	Beans and Containers		✓			
3	Spring Annotations			✓		
4	Aspect-Oriented Programming				✓	
5	Spring and Persistence					✓
6	Spring Web MVC and Spring Boot				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	1	1	1	-	-	-	-	-	-	2	2
CO2	3	2	3	2	-	-	-	-	-	-	2	1	2	2
CO3	3	2	3	2	2	1	-	-	-	2	2	3	3	3
CO4	3	2	3	2	2	2	-	-	1	-	2	1	3	1
CO5	2	2	3	3	2	-	1	1	-	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA869: HTTP Web Service for Enterprise Application

(200 Marks)

Contact Hours: 06

Pre-requisite: Work experience with C#, ASP.NET MVC, MSSQL, HTML, CSS, JavaScript and have some understanding of JQuery.

Methodology & Pedagogy: During theory lectures, illustrations emphasizing the need for advance features of WEB API and ASP.NET will be covered. During Practical sessions, students will require to develop Web API using concepts of .NET framework and other pre-requisite technologies, discussed during class.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	ASP.NET Web API - Introduction	4	36
2	MVC and Web API Controller and Entity Framework	8	
3	Web API – Routing	4	
4	Web API – Action Formatter and Filters	4	
5	Backend Validation and Exception Handling	8	
6	JQuery and AJAX	8	

Total Hours (Theory): 36
Total Hours (Lab): 36
Total Hours: 72

Detail Syllabus:

Unit I: ASP.NET Web API - Introduction **Hours 04**

Introduction to RESTFul WEB API, Characteristics of ASP.NET WEB API, versions of Web API, Difference among web service, Window Communication Foundation and WEB API, Test Web API Fiddler and Postman

Unit II: MVC and Web API Controller and Entity Framework **Hours 08**

Use of Web API Controller in Controller class of Web API, Functionality of Web API Controller, Difference between Web API controller and MVC controller, Action Method Naming Conventions, Action Result, MVC with WEB API, CRUD operation with Entity Framework in Web API.

Unit III: Web API – Routing **Hours 04**

Routing in Web API, Routing: Convention-based Routing and Attribute based Routing, Routing and Action Execution.

Unit IV: Web API – Action Formatter and Filters **Hours 04**

Data Formatter, Media Type Formatter, Web API Filter, Exception Filters: HttpResponseException, Exception Filters, Registering Exception Filters, HttpError

Unit V: Backend Validation and Exception Handling **Hours 08**

Model Validation with annotation, Custom Exception, recognizing need of custom Exceptions, Backend validation using Custom Exception for robustness and Data Integrity.

Unit VI: JQuery and AJAX **Hours 08**

Consume RESTFul Web API through GET, POST, PUT and DELETE. Understanding JQuery and JavaScript. JQuery usage and AJAX Request configuration and parameters understanding.

Core Books:

4. Mithun Pattankar, Malendra Hurbuns, Mastering ASP.NET Web API: Build powerful HTTP services and make the most of the ASP.NET Core Web API platform 1st Edition. – 2017.

Reference Books:

4. Jamie Kurtz, Brian Wortman, ASP.NET Web API 2: Building a REST Service from Start to Finish. - 2014

Web References:

1. https://www.tutorialspoint.com/asp.net_mvc/asp.net_mvc_web_api.htm

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Be familiar with RESTFul API and ASP .NET Web API.**
CO2 :	Be able to implement ASP .NET Web API Controller and configure its Routing and implementing Entity Framework.
CO3 :	Be able to Configure actions in API Controllers, Filters and Formatters.
CO4 :	Be able provide Data Integrity with validations with Exceptions.
CO5 :	Be able to call API through testing applications and AJAX calls from 3rd Party Applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	ASP.NET Web API - Introduction	√				
2	MVC and Web API Controller and Entity Framework		√			
3	Web API – Routing		√			
4	Web API – Action Formatter and Filters			√		
5	Backend Validation and Exception Handling				√	
6	JQuery and AJAX					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	3	3	1	3	-	-	2	2	2	2	2	2	2
CO2	2	1	2	2	3	2	2	3	-	3	2	2	3	3
CO3	2	2	2	2	3	2	2	2	-	1	2	2	3	2
CO4	3	3	3	3	2	3	2	2	-	2	3	2	3	2
CO5	2	3	3	1	3	2	2	3	2	2	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA870 : Software Quality Assurance (100 marks)

Contact Hours: 06

Pre-requisite: None

Methodology & Pedagogy: The theory sessions will cover the concept of software testing that can be applied to achieve quality of software. The practical sessions will cover the application of testing techniques at various levels of software development using manual and automated testing tools.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	**Practical**
1	Introduction to software quality and testing process	**05**	**36**
2	Unit level testing	**06**	
3	Integration testing	**07**	
4	System Test Design, Planning & Automation	**07**	
5	System Test Execution & Acceptance Testing	**06**	
6	Software Quality Assurance	**05**	

Total Hours (Theory): 36
Total Hours(Lab): 36
Total Hours: 72

Detailed Syllabus:

Unit – I: Introduction to software quality and testing process **Hours: 05**
Concept of Software Quality, Role of testing, Design test case: Stateless and State oriented test cases sources of Information for Test Case Selection, verification and validation, brief outline of levels of testing, overview of White Box, Black Box and Grey Box Testing.

Unit – II: Unit level testing **Hours: 06**
White box testing methods, concept of unit level testing, Introduction to Test case design techniques, Static Techniques: Informal Reviews, Walkthroughs, Technical Reviews, Inspection. Introduction to Dynamic Techniques. Control flow testing: Statement Coverage Testing, Branch Coverage Testing, Path Coverage Testing, Conditional Coverage Testing, Finite state machine(FSM) : Handling events and state transitions, Guard conditions, activities state, types of states, test coverage criteria.

Unit – III: Integration testing **Hours: 07**

Concept of Integration Testing, System Integration Techniques, Software and Hardware Integration, Test Plan for System Integration, Off-the-Shelf Component Integration, System level testing, types of system level tests, Black box testing methods: Equivalence Class Partitioning, Boundary Value Analysis, Decision Tables, Random Testing, Error Guessing, Category Partition.

Unit - IV: System Test Design, Planning & Automation **Hours: 07**

Test Design Factors, Requirement Identification, Characteristics of Testable Requirements, Test Design Preparedness Metrics, Test Case Design Effectiveness Structure of a System Test Plan, Introduction and Feature Description, Assumptions, Test Approach, Test Suite Structure, Test Environment, Test Execution Strategy, Test Effort Estimation, Scheduling and Test Milestones, System Test Automation, Evaluation and Selection of Test Automation Tools, Test Selection Guidelines for Automation, Characteristics of Automated Test Cases, Structure of an Automated Test Case, Test Automation Infrastructure

Unit - V: System Test Execution & Acceptance Testing **Hours: 06**

Preparedness to Start System Testing, Metrics for Tracking System Test, Metrics for Monitoring Test Execution, Beta Testing, First Customer Shipment, System Test Report, Product Sustaining, Measuring Test Effectiveness. Types of Acceptance Testing, Acceptance Criteria, Selection of Acceptance Criteria, Acceptance Test Plan, Acceptance Test Execution, Acceptance Test Report, Acceptance Testing in extreme Programming.

Unit - VI: Software Quality Assurance **Hours: 05**

Five Views of Software Quality, McCall's Quality Factors and Criteria, Quality Factors Quality Criteria, Relationship between Quality Factors and Criteria, Quality Metrics, Quality Characteristics, Software Quality ISO Standards. Elements of Software Quality Assurance, SQA Task, Goals and Metrics, Formal approaches to SQA, Statistical Software Quality Assurance, Software Reliability, SQA Plan.

Core Books:

1. Sagar Naik, Piyu Tripathy: Software Testing and Quality Assurance, Theory and Practice, Wiley, 2008.

2. Roger S Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, McGRAW HILL International Edition, 2010.

3.Lee Copeland: A Practitioner's Guide to Software Test Design, Artech House Publishers,2004.

Reference Books:

1. Boris Beizer: Software Testing Techniques: 2nd Edition, Van Nostrand Reinhold, 1990.

2. Daniel Galin: Software Quality Assurance, Pearson Education.

3. Ron Patton: Software Testing, Pearson Education, 2001.

Web References:

1. https://www.geeksforgeeks.org/software-engineering-white-box-testing/?ref=lbp [Introduction to Software Testing]

2. https://www.javatpoint.com/software-development-life-cycle [Software Development Life Cycle]
3. https://www.cs.ccu.edu.tw/~naiwei/cs5812/st7.pdf [Finite State Machine Testing]

Course Outcomes: Upon successful completion of the course, students will be

CO1 : **Able to understand basic concepts of software quality and testing**

CO2 : Familiar with concept of unit testing and methods of white box testing

CO3 : Able to apply black box testing methods

CO4 : Familiar with different concepts of software testing techniques

CO5 : Able to understand various standards of quality

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to software quality and testing process	√				
2	Unit level testing		√			
3	Integration testing			√		
4	System Test Design, Planning & Automation				√	
5	System Test Execution & Acceptance Testing				√	
6	Software Quality Assurance					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	3	3	2	2	2	1	2	3	3
CO2	3	2	2	2	2	3	3	3	3	2	3	2	3	3
CO3	2	2	3	2	2	2	2	2	2	3	1	2	2	2
CO4	3	3	3	2	3	3	3	1	3	3	3	2	2	1
CO5	3	2	3	2	1	3	2	2	2	2	2	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA871: Advanced Mobile Programming

(200 Marks)

Contact Hours: 06

Pre-requisite: Object Oriented Programming concepts

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and Cross Platform- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application using Dart language in Flutter. Student shall also develop applications with elegant user interface that deal with data storage using Firebase and state management

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to cross-platform development with Flutter and Dart	05	36
2	Creating UI with Flutter	07	
3	Building Apps with state	05	
4	Leveraging Flutter packages and structuring Flutter applications	08	
5	Incorporating backend data with Flutter application	05	
6	Integrating Flutter Application with Firebase and State Management	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Introduction to cross-platform development with Flutter and Dart **Hours 05**

Set up a new Flutter project using Android Studio. Widget tree, Interface design: pre-made Flutter Widgets. Image and Text Widgets. App Icons for iOS and Android. Add and load image assets to Flutter projects. Run Flutter apps on iOS Simulator, Android Emulator and physical devices.

Unit II: Creating UI with Flutter **Hours 07**

Hot Reload and Hot Restart, Use of Pubspec.yaml file, custom assets and fonts. An introduction to the Widget build(), layout widgets: Columns, Rows, Containers and Cards. Material icons, Icons class. Theme widgets. Refactoring widgets. Dart annotations and modifiers. Immutability of Stateless and Stateful Widgets. Update screen with the build(). Custom Flutter Widgets.Difference between final and const in Dart. Maps, enums and the ternary operator. Functions and arguments in Dart. Multi-screen Flutter apps, routes and the Navigator widget. Flutter favours composition vs. inheritance (customizing widgets).

Unit III: Building Apps with state **Hours 05**

About Stateful and Stateless Widgets, callbacks. Declarative style of UI programming, Flutter widgets react to state changes. Import Dart libraries. Variables, data types and functions in Dart. Build flexible layouts. Understand the relationship between setState(), State objects and Stateful Widgets.

Unit IV: Leveraging Flutter packages and structuring Flutter applications **Hours 08**

Dart package manager, Flutter compatible packages. The structure of the pubspec.yaml file. Incorporate the audio players package to play sound. Functions in Dart and arrow syntax. Refactor widgets, Flutter's philosophy of UI as code. The lists and conditionals in Dart. Classes and objects. Understand Object Oriented Dart. Dart Constructors for Flutter widgets. Design patterns to structure Flutter apps. Structuring and organizing Flutter apps.

Unit V: Incorporating backend data with Flutter application **Hours 05**

Asynchronous programming in Dart and use of async/await. Stateful Widget lifecycle methods, Handling exceptions in dart. Null aware operators. Location data from both iOS and Android. Http package and live data from open APIs. Parse JSON data using the dart:convert library. State objects via the Stateful Widget. Use the TextField Widget to take user input. Pass data backwards using the Navigator widget.

Unit VI: Integrating Flutter Application with Firebase and State Management **Hours 06**

Hero animations in Flutter apps. Animation controller, custom animations. The use of Dart mixins. Firebase Cloud Firestore into your Flutter apps. Authentication with Firebase Auth package. A scrolling ListView, reusable elements. Dart Streams: listen data changes. StreamBuild. Manage state across the widget tree. Declarative vs. imperative programming. setState, prop drilling and lifting state up. todo list app. The BottomSheet, ListViewBuilder. The Flutter app architecture design patterns. Manage state with the Google recommended Provider package.

Core Books:

1. Marco L. Napoli: Beginning Flutter: A Hands On Guide to App Development: Wrox publication: 2019.

Reference Books:

1. Eric Windmill: Flutter in Action: Edition: Manning Publication: January 2020.
2. Alessandro Biessek: Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart 2: Packt publication: September 2019

Web References:

1. https://docs.flutter.dev/reference/tutorials
2. https://www.tutorialspoint.com/flutter/index.htm
3. https://www.javatpoint.com/flutter
4. https://fluttertutorial.in/

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**able to clear all object oriented programming and cross platform concepts**
CO2 :	able to learn Flutter and Dart step by step
CO3 :	able to learn the reduce the code through native app performance, animated UI with material design and least testing
CO4 :	able to use Firebase to authenticate the users and use the remote database
CO5 :	able to build engaging native mobile apps for both Android and iOS

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to cross-platform development with Flutter and Dart	√				
2	Creating UI with Flutter	√	√			
3	Building Apps with state			√		
4	Leveraging Flutter packages and structuring Flutter applications			√		
5	Incorporating backend data with Flutter application					√
6	Integrating Flutter Application with Firebase and State Management				√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	3	2	1	3	-	1	-	1	-	-	1
CO2	3	1	3	2	3	1	2	1	1	-	1	1	1	2
CO3	3	-	3	1	3	1	1	2	1	-	1	1	1	2
CO4	3	3	3	3	3	1	1	3	1	-	2	2	3	1
CO5	3	3	3	1	3	1	1	3	1	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA872: Software Engineering with Agile and DevOps

(100 Marks)

Contact Hours: 04 hours

Pre-requisite: Analysis and design of Information Systems, Fundamental concepts of Object Oriented Programming.

Methodology & Pedagogy:

During theory sessions, the focus will be given on different concepts of software engineering and related aspects such as Project management and Agile software development. The emphasis will be given on the basics of software engineering concepts and UML. Students will get exposure to various concepts of DevOps, as well as Process Improvement and Reengineering.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to Software and Software Engineering	7
2	Requirement analysis and specification	8
3	Requirement Modeling using UML	8
4	Essence of Agile Development	9
5	Role of DevOps in SE	8
6	Software maintenance and risk management	8

Total Hours (Theory): 48

Total Hours:48

Detail Syllabus:

Unit I: Introduction to Software and Software Engineering **Hours 07**

The Role of Software in today's era, Software Engineering: A Layered Technology/software process layers, Software Process Models, The Linear Sequential Model, The Prototyping Model, Evolutionary Process Models

Introduction to software process: Generic process framework activities, Umbrella activities.

Unit II Requirement analysis and specification **Hours 08**

Understanding to System and Software Requirement, types of software requirements, Requirement Specification (SRS), Requirement Analysis and Requirement Elicitation, Requirement Engineering.

Introduction to the basic concepts of Project Scheduling, Project scheduling principles, relationships and effort distribution, Process Definition and Tailoring, Process Database and Process Capability Baseline, Effort Estimation and Scheduling, Project Management Plan, Configuration Management.

Unit III: Requirement Modeling using UML **Hours 08**

Introduction to UML, Structure modelling: Class Diagram and Behavioral Modelling. Under Behavioral Modelling: Use case Diagram, Interaction Diagram, Activity Diagram.

Unit IV: Essence of Agile Development **Hours 09**

Introduction to Agile Software Development, Characteristics of Agile Process, Agile methods, Principles of Agile methods, Agile Process models: XP, ASD, DSDM, Scrum, Crystal, FDD, AM.

Unit V: Role of DevOps in SE **Hours 08**

Overview, Problem Case Definition, Benefits of Fixing Application Development Challenges, DevOps Adoption Approach through Assessment, Solution Dimensions, What is DevOps?, DevOps Importance and Benefits, DevOps Principles and Practices, 7 C's of DevOps Lifecycle for Business Agility, DevOps and Continuous Testing, How to Choose Right DevOps Tools, Challenges with DevOps Implementation, Must Do Things for DevOps, Mapping My App to DevOps Assessment, Definition, Implementation, Measure and Feedback.

Unit VI: Software maintenance and risk management **Hours 08**

Software risks, risk identification, risk projection, risk mitigation, monitoring and management.
Introduction to software maintenance, types of maintenance.
Concept of software reengineering, Business Process Reengineering(BPR), BPR life cycle, reverse engineering, Applications of Software Engineering.

Core Books:

1. Roger S.Pressman, Software Engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Grady Booch, James Rambaugh, Ivar Jacobson: The Unified Modeling Language User
3. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.

Reference Books:

1. Ian Sommerville, Software engineering, Pearson education Asia
2. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
3. Merlin Dorfman (Editor), Richard H. Thayer (Editor), Software Engineering
4. Robert C. "Uncle Bob" Martin, Clean Architecture: A Craftsman's Guide to Software Structure and Design
5. Deepak Gaikwad, Viral Thakkar, DevOps Tools from Practitioner's ViewPoint, Wiley

Web References:

1. **https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology [Introduction to Software Engineering/Process/Methodology]**
2. https://www.uml-diagrams.org/ [UMLUnits]
3. https://nptel.ac.in/courses/106/101/106101061/ [Agile Software Development and Extreme Programming and Overview of DevOps]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Understanding Software Engineering Process Models**
CO2 :	Able to analyze requirements and mange project
CO3 :	Get familiar with the concepts of Unified Modeling Language
CO4 :	Understanding of the Agile software development concepts
CO5 :	Able to apply DevOps concepts and understand risk management

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction to Software Engineering	√				
2	Requirement analysis and project management		√			
3	Understanding UML			√		
4	Agile Development				√	
5	DevOps					√
6	Software maintenance and risk management					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	3	3	2	2	2	1	2	3	3
CO2	3	2	2	2	2	3	3	3	3	2	3	2	3	3
CO3	2	2	3	2	2	2	2	2	2	1	1	2	3	2
CO4	3	3	3	2	2	2	2	1	1	1	3	2	3	2
CO5	2	2	3	2	1	3	2	1	1	1	2	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS106.02 C: ACADEMIC WRITING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	**30/15**	--	30/15	**2**
Marks	**--**	100	--	**100**	

Pre-requisite courses:

- An Intermediate Guide to Writing in English for University Study https://www.futurelearn.com/courses/english-for-study-intermediate/4/todo/62943

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Academic Writing and Research Process	05
2.	Anatomy of Academic Writing	05
3.	Key Academic Skills	05
4.	Accuracy in Academic Writing	05
5.	Using and Citing Sources of Ideas	05
6.	Contemporary Practices in Academic Writing	05
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	**30**

Detailed Syllabus:

1.	Academic Writing and Research Process	5 Hours	
	Introduction to Academic Writing, Academic Writing as a Part of Research, Types of Academic Writing, Features of Academic Writing, Importance of Good Academic Writing in various Academic Works		
2.	Anatomy of Academic Writing	5 Hours	
	Academic Vocabulary, Simple and Complex Sentences, Organizing Paragraphs, The Writing Process, Adopting Academic Writing Style		
3.	Key Academic Skills	5 Hours	
	Note – taking, Note – making, Paraphrasing, Summarizing		
4.	Accuracy in Academic Writing	5 Hours	

	Lexical Range, Academic Language and Structures, Elements of Writing, Proof Reading, Editing, and Rewriting		
5.	Using and Citing Sources of Ideas	5 Hours	
	Academic Texts and their Types, Intellectual Honesty in Academic Writing, Avoiding Plagiarism – Idea Theft, Degrees of Plagiarism, Types of Borrowing, Anatomy of Citations, Common Citation Styles		
6.	Contemporary Practices in Academic Writing	5 Hours	
	Analytical Essays, Graph / Table / Process Interpretation and Description, Writing Reports and Abstract, Writing Research / Concept Papers		

Course Outcome (COs):

At the end of the course, the students

CO1	Will have sound understanding of the concept and applications of academic writing
CO2	Will have acquired enough knowledge of academic writing style, strategy and approach
CO3	Will be able to demonstrate error free and effective academic writing
CO4	Will be able to demonstrate ability to work on project/report/paper writing
CO5	Will have the sound understanding of the Research and Research Methodology
CO6	Will be effectively communicating in diverse academic and professional settings.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	3	-	3	-	-	-	-	2
CO6	-	-	-	-	-	-	2	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- **Text book:**
  1. Academic Writing for International Students, Routledge
  2. Academic Writing: A Guide for Management Students and Researchers. Monipally, M.M. &Pawar, B.S. Sage. 2010. New Delhi
  3. Effective Academic Writing Level - 1,2,3,4 (Second Edition) By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

- **Reference book:**
  1. Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
  2. Development Communication In Practice by Vilanilam V J, Sage
  3. Intercultural Communication by Mingsheng Li, Patel Fay, Sage

- **Web material:**

  www.owl.perdue.edu

DETAILED SYLLABUS

FOR

MCA PROGRAMME

(3rd SEMESTER)

EFFECTIVE FROM

ACADEMIC YEAR 2022-23

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA934-937	Elective-III	3	3	6	6	10	20	70	15	15	70	200
CA938	Advanced Web Designing	3	3	6	6	10	20	70	15	15	70	200
CA939	Mini Project	-	12	12	12	-	-	-	100		300	400
CA940	Green Computing		2	2	02	-	-	-	30		70	100
		06	20	26	26	200			700			900

Elective-III	
Course Code	**Course Title**
CA934	Digital Image Processing
CA935	Object Oriented Programming for Smart Contracts
CA936	Data Analytics
CA937	Internet of Things

CA934: Digital Image Processing

(100 Marks)

Contact Hours: 03

Pre-requisite: Basic knowledge of programming

Methodology & Pedagogy: The fundamental concepts of digital image processing should be covered in the beginning lectures. Various image enhancement and colour image processing techniques should be explained in details. The methods of image segmentation, representation and compressions should be taught in-depth.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Digital Image Fundamentals	05	
2	Colour Image Processing	07	
3	Image Enhancement	05	36
4	Image Segmentation	07	
5	Image Representations and Recognition	06	
6	Image Compressions	06	

Total Hours (Theory): 36

Total Hours (Lab) : 36 Hours

Total Hours: 72 Hours

Detail Syllabus:

Unit I: Digital Image Fundamentals **Hours 05**

Introduction, Origin, Component of image processing system, Human visual system, Image as a 2D data, Image representation, Gray scale and Colour images, image sampling and quantization, Basic Relationships between Pixels.

Unit II: Colour Image Processing **Hours 07**

Colour fundamentals, Colour models, Colour transformation, Smoothing and Sharpening, Colour segmentation, Noise Models, Noise Reduction, Inverse Filtering.

Unit III: Image Enhancement **Hours 05**

Basic gray level Transformations, Histogram Processing, Spatial Filtering, Preliminary Concepts, Extension to functions of two variables, Image Smoothing, Image Sharpening, Frequency domain filtering, Homomorphic filtering

Unit IV: Image Segmentation **Hours 07**

Edge based segmentation, Region based segmentation, Region split and merge techniques, Region growing by pixel aggregation, optimal thresholding, Hough Transform, boundary detection.

Unit V: Image Representation and Recognition **Hours 06**

Boundary representation, Chain code, Polygonal Approximation, Signature, Boundary segments, Boundary description, Shape Number, Regional Descriptors, Topological Feature, Texture, Pattern and Patterns classes, Recognition Based on matching

Unit VI: Image Compression **Hours 06**

Introduction, Image compression model, Error free compression, Variable length coding, Bit plan coding, Lossless predictive coding, Lossy compression, Compression standards.

Core Books:

5. Rafael C. Gonzalez, Richard E. Woods: Digital Image Processing, Fourth edition, Pearson education, 2017
6. Anil K. Jain: Fundamentals of Digital Image Processing, First Edition, Pearson Education, 2015

Reference Books:

1. Burger, Wilhelm, Burge, Mark J: Principles of Digital Image Processing, Springer, 2009.
2. Milan Sonka, Vaclav Hlavac, Roger Boyle: Image Processing, Analysis, and Machine Vision, CL Engineering, 2007

Web References:

1. https://www.tutorialspoint.com/dip/ [Basics of Image Processing]
2. https://sisu.ut.ee/imageprocessing/book/1 [All concepts of digital image processing]
3. https://www.cs.auckIand.ac.nz/courses/compsci773s1c/lectures/ImageProcessing-html/topic3.htm [Image segmentation]
4. http://www.eIeteI.p.Iodz.pI/mstrzeI/imageproc/enhancement1.PDF [Image enhancement]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand the concept of Explain various types of images and analyze various techniques for intensity transformation and spatial filtering with applications.
CO2 :	Understand the concepts of colour image processing and noise models.
CO3 :	Understand the concepts of image enhancement and its techniques.

CO4 :	Learn various image segmentation techniques.
CO5 :	Be able to represent and recognize images using various pattern recognition methods its compression techniques.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Digital Image Fundamentals	√				
2	Colour Image Processing		√			
3	Image Enhancement			√		
4	Image Segmentation				√	
5	Image Representations and Recognition					√
6	Image Compressions					√

Course Articulation Matrix:

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	3	3	2	1	1	1	1	-	1	1	3	2	3
CO2	3	3	3	2	1	1	1	1	-	1	1	3	2	3
CO3	3	3	3	2	1	1	1	1	-	1	1	3	2	3
CO4	3	3	3	2	1	1	1	1	-	1	1	3	2	3
CO5	3	3	3	2	1	1	1	1	-	1	1	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA935: Object Oriented Programming for Smart Contracts

(200 Marks)

Contact Hours: 06

Pre-requisite: CA861:Blockchain Technology

Methodology & Pedagogy: During the lecture sessions, the students will learn about fundamental of smart contracts and related applications. Practical session will provide hands-on in order to implement smart contracts using Solidity programming language. Practical sessions will cover basics of Solidity, Expression and Control Statements, Object Oriented constructs and Testing, Debugging and Deployment. Through case study approach, students will explore smart contracts in real applications.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Smart Contracts	6	36
2	Basics of Solidity	7	
3	Expression & Control Statements	7	
4	OOPs Construct in Solidity	7	
5	Testing, Debugging and Deployment	5	
6	Smart contracts in Real World	4	

Total Hours (Theory): 36 Hours

Total Hours (Lab) : 36 Hours

Total Hours: 72 Hours

Detail Syllabus:

Unit I : Introduction to Smart Contracts **Hours 06**

An overview to the history of smart contracts, An intro to the life-cycle of a smart contract, Ethereum's smart contract languages, Interfacing with Ethereum Networks (overview of Ethereum Networks, Clients, Wallets, Transactions etc.), The Solidity Programming Language, Development Environments

Unit II : Basics of Solidity **Hours 08**

Overview of Solidity, Solidity-Environment Setup, Basic Syntax, Smart contract layout, The structure of .sol source file, First Application, Understanding the different compiler versions and pragmas, Authoring smart contracts, Contract definitions, Basic data types, Variables, Variable Scope, Operators, Structs and Enums, Mapping and Arrays, Build-in Functions, User Functions

Unit III : Expression & Control Statements **Hours 06**

Expressions and Control Structures, Valid expressions of the language, Conditional logic, Implementation of loops, Exception Handling, Events and Logging

Unit IV : OOPs Construct in Solidity **Hours 07**

Contract constructor and self-destruct, Function Modifiers and Fallback functions, calling other contracts, Inheritance and Multiple Inheritance, Declaring Abstract Classes and Interfaces, Implementation of Abstract interfaces, Function Overloading

Unit V : Testing, Debugging and Deployment **Hours 05**

Smart Contract testing, Basics of using Truffle for testing, types of testing in smart contract, Importance of testing in smart contracts, Debugging, Smart Contract Security – overview of attacks on Ethereum smart contracts Deployment.

Unit VI : Smart contracts in Real World **Hours 04**

Smart contracts in trading, investing, gaming, healthcare and real estate

Core Books:

1. Antonopoulos, Andreas M., and Gavin Wood: Mastering ethereum: building smart contracts and dapps : Edition (1st, 2nd , etc.) : O'Reilly Media: 2018.

Reference Books:

1. Ritesh Modi: Solidity Programming Essentials: A guide to building smart contracts and tokens using the widely used Solidity language: Edition (2nd) : Kindle Edition : Year of Publication.
2. Wei-Meng Lee: Beginning Ethereum Smart Contracts Programming: With Examples in Python, Solidity, and JavaScript: Apress: 2019

Web References:

1. https://www.blockchain-council.org/blockchain/smart-contracts-works/ [Smart Contracts Introduction]
2. https://101blockchains.com/solidity-tutorial/ [Solidity Introduction]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Able to understand the fundaments of Smart Contracts.**
CO2 :	Able to learn basics and object oriented construct of Solidity programming language.
CO3 :	Able to implement smart contracts using Solidity programming language.
CO4 :	Able to test, debug and deploy smart contract.
CO5 :	Able to understand use cases of smart contract in real world.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Smart Contracts	√				
2	Basics of Solidity		√	√		
3	Expression & Control Statements		√	√		
4	OOPs Construct in Solidity		√	√		
5	Testing, Debugging and Deployment				√	
6	Smart contracts in Real World					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	1	2	-	1	3	-	-	-	-	2	1	3
CO2	3	1	2	1	3	-	3	-	-	-	-	2	2	3
CO3	3	1	2	1	3	-	3	-	-	-	-	2	2	3
CO4	3	1	3	2	3	-	3	-	-	-	-	2	3	3
CO5	1	1	1	2	1	-	3	-	-	2	2	2	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA936: Data Analytics

(200 Marks)

Contact Hours: 06

Pre-requisite: CA845: Advanced Database Technologies

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of data analytics and related tools and techniques. Students will be introduced to the concepts of data science, exploratory data analysis, supervised and unsupervised learning methods. Applications as well as research trends and future direction of data analytics will be discussed with the length. During the practical sessions, students will be introduced to tools of data analytics such as R and Python. Students will be given appropriate case studies of data analytics to get the real time exposure of data analytics.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Theory Hours	Practical
1	Introduction of Data Analytics	06	36
2	Exploratory Data Analysis	06	
3	Supervised & Unsupervised Learning Methods	06	
4	Deep Re-enforcement Learning	06	
5	Data visualization	06	
6	Recent trends and future directions	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detail Syllabus:

Unit I : Introduction of Data Analytics Hours 06

What is Data Science? ,Big Data and Data Science hype ,Why now? , Datafication , Current landscape of perspectives , Skill sets needed, Data Analysis cycle, Data analysis Vs Data analytics, Life cycle of data

Unit II : Exploratory Data Analysis Hours 06

Philosophy of EDA - The Data Science Process ,Statistical Inference , Populations and samples , Statistical modeling, probability distributions, fitting a model Introduction to R ,Basic tools (plots, graphs and summary statistics) of EDA, SQL with data life cycle, Python for EDA.

Unit III : Supervised & Unsupervised Learning Methods Hours 06

Supervised methods: Linear Regression , Classification Trees, Random Forest, Clustering, Support Vector Machine, Regularization, Radial Basis, Association Rule Mining, Dimensionality Reduction - Singular Value Decomposition – Principal Component Analysis

Unit IV : Deep Re-enforcement Learning Hours 06

Introduction to Neurons, Activation functions, single and multi-layer perceptron, back propagation, Application, Hyper-parameters and Estimation, Gradient Descent, Curiosity driven learning.

Unit V : Data visualization Hours 06

Introduction of data visualization, Power BI, and Tableau, ggplot+ggplot2, Seaborn.

Unit VI : Recent trends and future directions Hours 06

Data Visualization - Basic principles, ideas and tools for data visualization, Data Science and Ethical Issues - Discussions on privacy, security, ethics , Next-generation data scientists, Basics of Big Data analytics,

Core Books:

1. Cathy O'Neil and Rachel Schutt: Doing Data Science, Straight Talk From The Frontline, O'Reilly. 2014.
2. Jure Leskovec, Anand Rajaraman, and Jeffrey David Ullman,Mining of Massive Datasets Cambridge University Press,2nd Edition, New York, NY, USA,2014.
3. Howard B. Demuth, Mark H. Beale, Orlando De Jess, and Martin T. Hagan, Neural Network Design , paperback USA, 2nd Edition,2014.

Reference Books:

1. 1. Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. , Probability & statistics for engineers & scientists ,9th edition, Prentice Hall,2012.
2. 2. Haykin, S. S., Haykin, S. S., Haykin, S. S., & Haykin, S. S., Neural networks and learning machines Pearson, Vlolume 3,2009.
3. 3. Mohammed J. Zaki and Wagner Miera Jr, Data Mining and Analysis: Fundamental Concepts and Algorithms.,Cambridge University Press. 2014.

Web References:

1. https://onlinecourses.nptel.ac.in/noc17_mg24/preview [Online Data Analytics Course]
2. https://www.itl.nist.gov/div898/handbook/eda/section1/eda11.htm [Exploratory Data AnalysisMaterial]
3. https://datahoarder.io/Humble%20Bundle%20Books/Humble%20Book%20Bundle_%20Data%20Science%20presented%20by%20O_Reilly/doingdatascience.pdf [Data Science E-Book]
4. http://www.astro.caltech.edu/~george/aybi199/Donalek_Classif.pdf [Supervised andunsupervised methods tutorials]
5. https://yourstory.com/2017/12/data-analytics-future-trends/ [Future Trends of Data Analytics]
6. http://www.astro.caltech.edu/~george/aybi199/Donalek_Classif.pdf [Supervised andunsupervised methods tutorials]
7. https://yourstory.com/2017/12/data-analytics-future-trends/ [Future Trends of Data Analytics]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Describe what Data Science and Data Analytics are and the skill sets needed to be a data scientist.
CO2 :	Understand significance of exploratory data analysis in statistical and visualization aspects.
CO3 :	Understand and apply data analytics techniques such as supervised, unsupervised and EDA.
CO4 :	Understand the importance of self-learning environments
CO5 :	Able to understand the recent trends and future directions of data analytics.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction of Data Analytics	√				
2	Exploratory Data Analysis		√			
3	Supervised & Unsupervised Learning Methods			√		
4	Deep Re-enforcement Learning				√	
5	Data visualization				√	
6	Recent trends and future directions					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	-	-	1	-	-	-	2	3	1
CO2	2	2	1	1	-	1	-	-	-	-	-	2	2	1
CO3	1	3	3	2	-	1	3	-	-	-	-	2	1	1
CO4	2	1	2	2	2	1	2	-	-	-	-	2	1	2
CO5	1	1	2	3	1	-	2	-	-	-	-	3	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA937- Internet of Things

(200 Marks)

Contact Hours: 06

Pre-requisite: - None

Methodology & Pedagogy: In order to achieve the objectives and goals, students will be taught the basics of IOT with its structure, layers and applications. Fundamentals controllers are introduced which can be used to implement the IOT based projects. Students will also be introduced to interface GPIO with controllers like arduino and Raspberry PI. They can also learn to develop a desktop application and mobile application which can control the device remotely.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Internet of Things	06	36
2	Programming Internet of Things System	08	
3	Sensors and Actuators for Internet of Things System	04	
4	Designing Mobile Application and Webpage for Internet of Things System	06	
5	ESP8266 and ESP32 Microcontroller for IOT	06	
6	Raspberry Pi for IOT Projects	06	

Total Hours (Theory): 36

Total Hours (Lab) :36

Total Hours:72

Detail Syllabus:

Unit I : Introduction to Internet of Things : Hours: 06

Introduction - Overview of Internet of Things (IoT), The building blocks of IoT, Various technologies making up IoT ecosystem, IoT levels, IoT design methodology, The Physical Design and Logical Design of IoT, Functional blocks of IoT and Communication Models, Development Tools used in IoT.

Unit II : Programming Internet of Things System: Hours: 08

Arduino / node MCU controller for IOT Programming, Embedded C data types, variables, constants and operators, programming using control statements, loops, string and array, Arduino IDE. Variables and Numbers, Looping Structures, Conditional Statements, Lists, Tuples and Dictionaries, Type Conversions, Function declaration, calling functions and passing values, Function Returning values. Exiting from functions.

Unit III : Sensors and Actuators for Internet of Things System: Hours: 04

Interfacing with various sensors like temperature sensor, PIR sensor and ultrasonic sensor. Interfacing with servomotor, DC motor, Gas sensor.

Unit IV : Designing Mobile Application and Webpage for Internet of Things System Hours: 06

Basic concepts of mobile application development, designing webpage for IoT user, Connecting App / webpage with cloud and controller as per Internet of Things system levels.

Unit V : ESP8266 and ESP32 Microcontroller for IOT Hours: 06

Basics of Wireless Networking, Introduction to ESP8266 Wi-Fi Module, Various Wi-Fi library, Web server- introduction, installation, configuration, Posting sensor(s) data to web server, Introduction to ESP32 microcontroller.

Unit VI : Raspberry Pi for IOT Projects Hours: 06

Introduction to Raspberry Pi, Why Raspberry pi, Features of Raspberry pi, Basic set up and first boot configuration, Different uses of Raspberry pi, Different Versions of Raspberry pi, Vision of Raspberry pi, Different components of the Board.

Core Books:

1. Arshdeep Bahga, Vijay Madisetti: Internet of Things: A Hands-On Approach, VPT Publication, 2014.
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

Reference Books:

1. M. Richardson, S. Wallace: Getting Started with Raspberry Pi, O'Reilly, 2012
2. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication

Web References:

1. https://www.tutorialspoint.com/arduino/index.htm [Arduino Fundamentals]
2. www.tinkercad.com [Arduino Simulator]
3. https://www.youtube.com/watch?v=UUOCh0Cbty8[Raspberry Pi - How to start programming with Python]
4. https://www.raspberrypi.org/documentation/usage/python/[Step by Step Programming]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Ability to understand IOT Terminology**
CO2 :	Understanding IOT Programming
CO3 :	Understanding use of sensors and actuators
CO4 :	Learning Mobile app for IOT Projects
CO5 :	Designing IOT projects using ESP8266 and ESP32 and Raspberry PI

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction to Internet of Things	√				
2	Programming Internet of Things System		√			
3	Sensors and Actuators for Internet of Things System			√		
4	Designing Mobile Application and Webpage for Internet of Things System				√	
5	ESP8266 and ESP32 Microcontroller for IOT					√
6	Raspberry Pi for IOT Projects					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	3	1	1	1	1	1	1	-	1	-	-

CO2	3	1	2	3	1	1	1	2	2	-	-	1	1	-
CO3	2	2	2	1	1	1	2	2	2	-	-	2	1	1
CO4	2	2	3	2	1	2	2	2	2	-	-	2	1	1
CO5	1	2	3	2	1	2	3	2	2	-	-	2	1	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA938: Advance web Designing

(200 Marks)

Contact Hours: 06

Pre-requisite: Working knowledge of HTML and JavaScript

Methodology & Pedagogy: During the theory sessions students will understands the various JavaScript frameworks and its architecture, also able to comprehend MEAN stack application concepts. During the practical sessions students will learn how to reduce the amount of code you write to build rich user interface applications, Modularizing code, retrieving data from back-end-server and manipulate it.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to JavaScript Frameworks	04	03
2	Directives in AngularJS	08	06
3	MVW-The AngularJS way	07	06
4	Introduction to NodeJS	06	09
5	Working with NodeJS Framework - Express	06	06
6	Working with Database	05	06

Total Hours (Theory): 36

Total Hours (Lab) :36

Total Hours:72

Detail Syllabus:

Unit I: Introduction to JavaScript Frameworks. **Hours 04**

JavaScript Frameworks & Libraries, MEAN.JS introduction, Architecture of MEAN.JS, Introduction to AngularJS, Advantages of AngularJS, Data Binding in angularJS.

Unit II : Directives in AngularJS **Hours 08**

Working with Expressions in angularJS, Introduction to Directives, Directive Lifecycle, Using Angular JS built-in directives, creating a custom directive, Overview of $scope - lifecycle of $scope, $rootscope, $watch, $apply.

Unit III : MVW-The Angular JS way. **Hours 07**

MVW Architecture: Model-View-Controller and Model-View-View-Model Architecture, Introduction to AngularJS Modules – Application module & Controller modules, Attaching Properties and functions to scope, Controller in external files, AngularJS Filters, working with Angular Forms, Form events, validating Angular forms.

Unit IV : Introduction to NodeJS **Hours 06**

Introduction to NodeJS, Advantages of NodeJS, working with Node Package Manager, Installing Modules using NPM, Traditional web Server Model, Node.js process Model, NodeJS modules – Core Modules, Local Module, Third-party modules, Export Module in Node.js, Creating NodeJS web server.

Unit V : Working with NodeJS Framework - Express **Hours 06**

Introduction to ExpressJS, Advantages of ExpressJS, Installing Express.js, building your first web server, Serve Static Resources using Express.js, working with HTTP methods of ExpressJS, ExpressJS routing.

Unit VI : Working with Database **Hours 05**

Introduction to Mongo DB, Basic operations using MongoDB, Access MongoDB in Node.js, setting up mongoose, Connecting MongoDB, Insert, update and delete document.

Core Books:

1. Jeffry Houser : "Learn With: Angular 5, Bootstrap, and NodeJS", Kindle Edition,
2. Shyam Seshadri Brad Green:"AngularJS – Up and Running, Brad Green", Second Edition, O'REILLY
3. Basarat Ali Syed: "Beginning Node.js", Apress Publication.
4. Greg Lim : "Beginning MEAN Stack (MongoDB, Express, Angular, Node.js)", kindle Edition.

Reference Books:

1. Agus Kurniawan: "AngularJS Programming by Example 2017 Edition", Kindle Edition.
2. Adam Freeman: " Pro AngularJS 2017 Edition", Apress.
3. Krasimir Tsonev: "Node.js By Example", Packt Publishing
4. Ethan Brown: "Web Development with Node and Express", O'REILLY

Web References:

1. http://www.w3schools.com/angular/default.asp [Tutorial link for AngularJS]
2. http://www.tutorialspoint.com/angularjs/ [Tutorial link for AngularJS]
3. https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf [E-book for AngularJS]
4. http://www.tutorialsteacher.com/nodejs/nodejs-modules [Tutorial link for NodeJS]
5. https://www.javatpoint.com/mean-stack-tutorial [MEAN stack development]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Familiar with JavaScript frameworks.**
CO2 :	Able to understand basic fundamental of AngularJS.
CO3 :	Able to implement Model-View-View Model architecture.
CO4 :	Able to create web server.
CO5 :	Able to work with database.
C06:	Able to create Single page application and MEAN stack application.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**					
		CO1	**CO2**	**CO3**	**CO4**	**CO5**	**C06**
1	Introduction to JavaScript Frameworks	√					√
2	Introduction to AngularJS		√				√
3	MVW-The Angular JS way			√			√
4	Introduction to NodeJS			√	√		√
5	Introduction to ExpressJS			√	√		√
6	Working with Database					√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	1	1	2	-	-	1	1	1
CO2	-	2	2	1	3	-	1	2	-	1	-	1	1	1
CO3	2	2	3	3	3	2	2	3	2	2	2	2	3	3
CO4	-	1	3	1	3	1	2	1	1	1	-	1	2	1
CO5	1	1	3	1	3	1	2	2	-	1	-	1	2	1
C06	3	3	3	3	3	2	3	3	2	2	2	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “

CA939: Mini Project

(300 Marks)

Contact Hours: 12

Objective:

The main objective of this mini project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Course Outcomes:

C01: Student will understand the implementation of concepts of SDLC and Software Engineering.

C02: The programming concepts they learn during their academics, it will be converted in to the actual implementations.

C03: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

C04: Students must understand the difference between a program and professional application/product/software.

C05: Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Mini Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Mini Project to the institute. Each Student is required to make a copy of Mini Project in CD and submit along with Mini Project report.

Evaluation:

The project must be evaluated in two aspects:

a. Internal (100 Marks):
    i. Reporting to internal project guide
    ii. Incorporation of suggestions by project guide
    iii. Internal Project viva examination
b. External (300 Marks):

i.Project Report Preparation & Evaluation
ii.External Project Viva Examination

Course Code	**Course Title**	**Teaching Scheme**		**Internal**	**End Semester Examination**		**Total**
		Contact Hours	**Credit**	**Continuous Evaluation**	**Report**	**Presentation & Viva**	
CA939	Mini Project	12	12	100	100	200	400

Web References:

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf[Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%2 0Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf[ForEffective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf[For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

CA940: Green Computing
(100 Marks)

Contact Hours: 02

Pre-requisite: None

Methodology & Pedagogy: During practical sessions, the basic concepts regarding green computing, green framework and green compliance will be discussed. The core concepts towards green computing will be covered so that students can analyze and understand various aspects related with green computing.

Outline of the course:

Week No	Practical	Description
1 - 2	**Carbon footprint of Green Computing**	• Analysis and survey about the carbon footprint of campus. • Measuring the carbon footprint, a small MIDC Company. • Carbon footprint for FMCG companies • CORPORATE CARBON FOOTPRINT AND CARBON NEUTRALITY. • Measures to reduce carbon footprint
3 - 4	**Energy Conservation in Green Computing**	• Study on Green Assets (Buildings, Data Centers, Networks, and Devices, Green Information Systems) Design and Development Models. • Plan to Cut Down Your Electricity Bill • Measuring and monitoring use of Power in XXX Company. • A survey of steps taken to be energy efficient by Malls / Shopping centers • Energy Efficient cooling solutions for homes and offices • Electricity Smart Grids and smart energy systems
5 - 6	**Recycling in Green Computing**	• Recycling of IT waste in Colleges • Recycling vs Reuse • Recycling initiatives taken up by XXX Housing Society: A Case Study

		• Damage caused by improper recycling of e-waste in developing countries like India or China
7 - 8	**Going Paperless and Datacenters in Green Computing**	• A Case study of a traditional company going paperless with the use of Electronic media • Going paperless in Government Departments • Challenges in going paperless in Indian Context • Economic benefits of going Paperless • Survey of best energy-efficient practices in data centers around the world • Designing a datacentre with use of green technology. • Design considerations of datacentres for efficient cooling • Impact of Datacenters on the environment
9 - 10	**Review of Green Initiatives in India and abroad**	• Submitting a small research study on National Mission for a Green India (GIM). • Submitting a small research study on Indian Green Building Council (IGBC). • Submitting a small research study on The Indian Council of Agricultural Research. • Submitting a small research study on LEED INDIA. • Basel Convention on the Control of Transboundary Movements of Hazardous Waste and their Disposal. • Initiatives taken by Europe to reduce the toxic effect of e-waste on environment • E Parisaraa - India's First E Waste. • WEEE Recycle India - E-Waste Collection Centres in Bangalore.

		• E-waste management rules implemented by Ministry of Electronics & Information Technology by Government of India.
11 - 12	**Creating Prototype model**	• Build a prototype/Working model of a wind turbine at home. • Build a prototype/Working model of a simple solar equipment's. • Build a prototype/Working model of Water Purification System. • Rain water harvesting model preparation. • Creating greywater recycling system. • Paper bag making and distributing.

Total Hours (Practical): 24
Total Hours:24

Core Books:

1. Bhuvan Unhelkar: "Green IT Strategies and Applications-Using Environmental Intelligence", CRC Press, June 2014.

2. Woody Leonhard, Katherine Murray: "Green Home computing for dummies", August 2012.

Reference Books:

1. Alin Gales, Michael Schaefer, Mike Ebbers: "Green Data Center: steps for the Journey", Shroff/IBM rebook, 2011.

2. John Lamb: "The Greening of IT", Pearson Education, 2009.

3. Jason Harris: "Green Computing and Green IT- Best Practices on regulations & industry", Lulu.com, 2008.

4. Carl speshocky: "Empowering Green Initiatives with IT", John Wiley & Sons, 2010.

5. Wu Chun Feng : "Green computing: Large Scale energy efficiency", CRC Press

Web References:

1.https://dataconomy.com/2022/06/what-is-green-computing-definition/
2.https://www.techrockstars.com/the-essential-guide/what-is-green-computing/
3.https://www.smallbusinesscomputing.com/guides/green-computing/

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Can understand the fundamentals and Carbon footprint of Green Computing
CO2 :	Can analyze and understand Green assets and its importance
CO3 :	Can analyze the Green computing Grid Framework.
CO4 :	Can understand the issues related with Green compliance.
CO5 :	Can study and develop various applications and case studies.

Course Outcomes Mapping:

Week No.	**Practical Title**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1-2	Green IT Fundamentals and Carbon footprint of Green Computing	√				
3-4	Green Assets and Energy Conservation in Green Computing		√			
5-6	Recycling and conservation		√			
7-8	Going Paperless and Datacenters, Green computing frameworks			√		
9-10	Review of Green Initiatives in India and abroad				√	
11-12	Green Modeling, prototype and applications					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	1
CO2	-	2	-	-	-	2	-	-	-	-	2	-	2	2
CO3	-	2	2	2	-	2	2	-	-	2	2	-	-	2

CO4	-	-	-	-	2	3	-	-	-	-	-	-	-	2
CO5	-	-	-	-	-	1	-	-	1	1	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

DETAILED SYLLABUS

FOR

MCA PROGRAMME
(4th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2022-23

Teaching & Examination Scheme

Master of Computer Applications (M.C.A.) Programme

(Choice Based Credit System)

Effective from Year 2022 - 23

Semester-IV

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
CA941	Dissertation/ Project Work	2	28	30	30	200	200	400	800

CA941: Dissertation/Project Work

(800 Marks)

Contact Hours: 30

Objective:

The main objective of this Dissertation is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Course Outcomes:

C01:	Student will understand the implementation of concepts of SDLC and Software Engineering.
C02:	The programming concepts they learn during their academics, it will be converted in to the actual implementations.
C03:	Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
C04:	Students must understand the difference between a program and professional application/product/software.
C05:	Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Dissertation is in house project development. Every student is required to carry out Dissertation work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Dissertation work during the semester as per the schedule provided by the placement Coordinator.

Dissertation proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Dissertation to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Dissertation to the institute. Each Student is required to make a copy of Dissertation in CD and submit along with Dissertation report.

Evaluation:

The project must be evaluated in two aspects:

a. Internal (200 Marks):
    i. Reporting to internal project guide
    ii. Incorporation of suggestions by project guide

iii. Internal Project viva examination

b. External (600 Marks):

i. Project Report Preparation & Evaluation

ii. External Project Viva Examination

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Web References:

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf[Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%2 0Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf[ForEffective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Templa te .pdf[For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation